

模拟电子技术基础

Analog Device

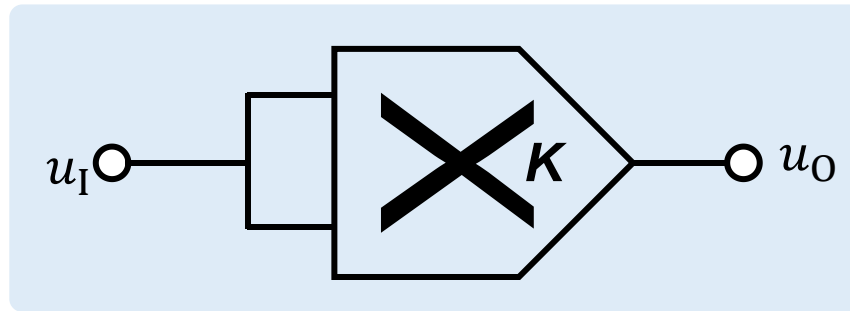
8 Signal Processing

信号的处理

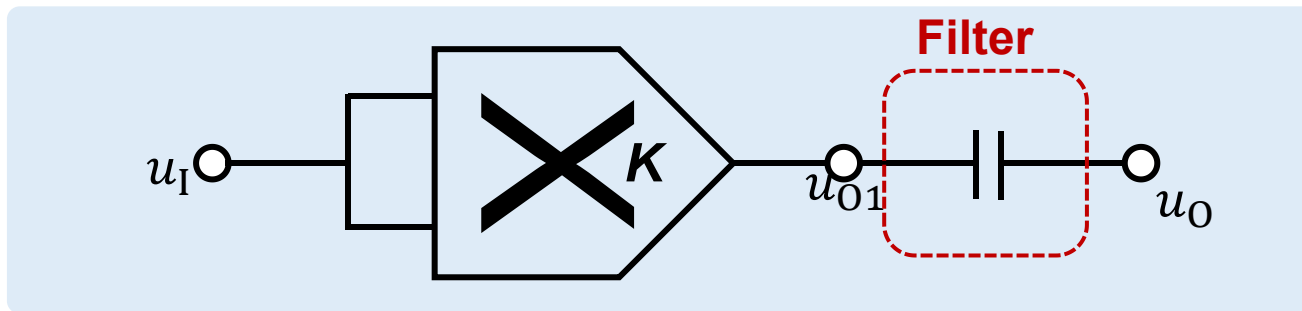
陈晓龙 Chen, Xiaolong

电子与电气工程系 E³

8.1 Filters 滤波

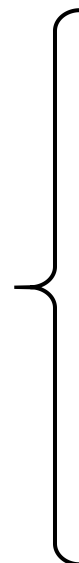


$$u_I = \sqrt{2}U_i \sin \omega t \quad u_O = KU_i^2 (1 - \cos 2\omega t)$$



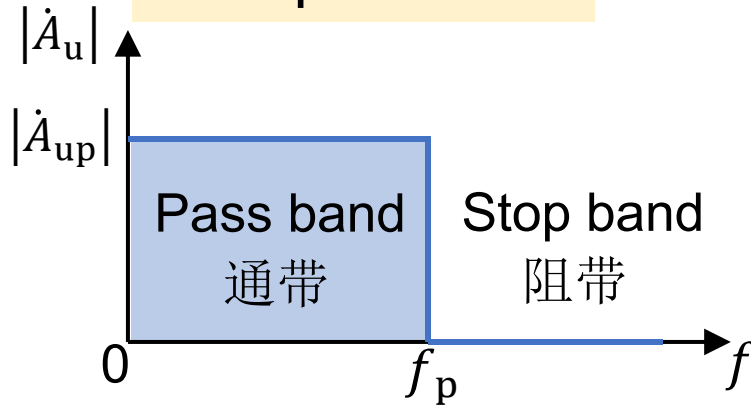
$$u_O = -KU_i^2 \cos 2\omega t$$

Based on the operating frequency:

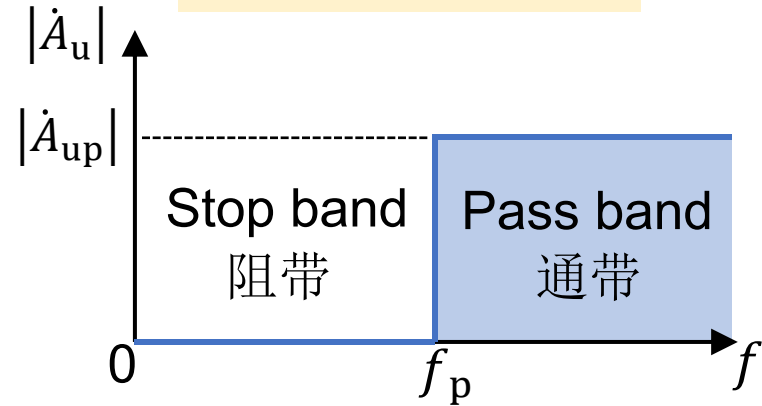
- 
- Low pass filter 低通滤波
 - High pass filter 高通滤波
 - Band pass filter 带通滤波
 - Band Elimination filter 带阻滤波

Ideal filters 理想滤波器

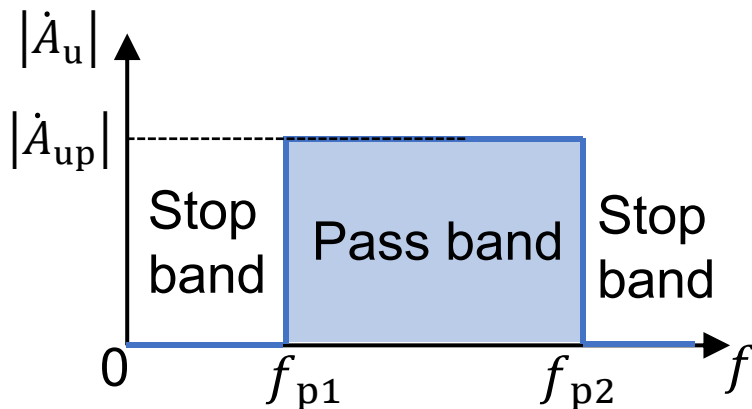
Low pass filter



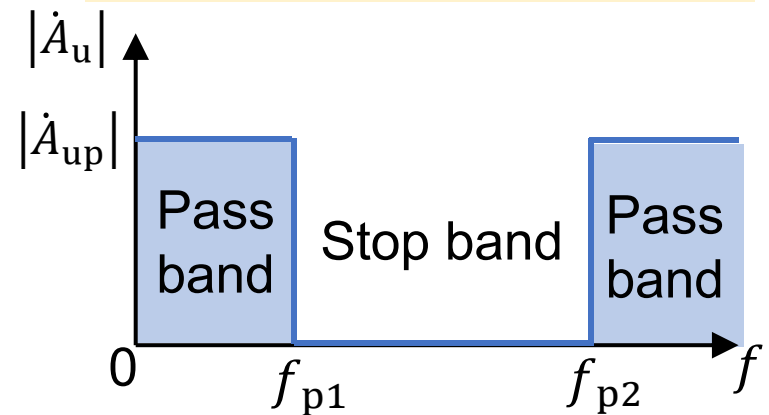
High pass filter



Band pass filter

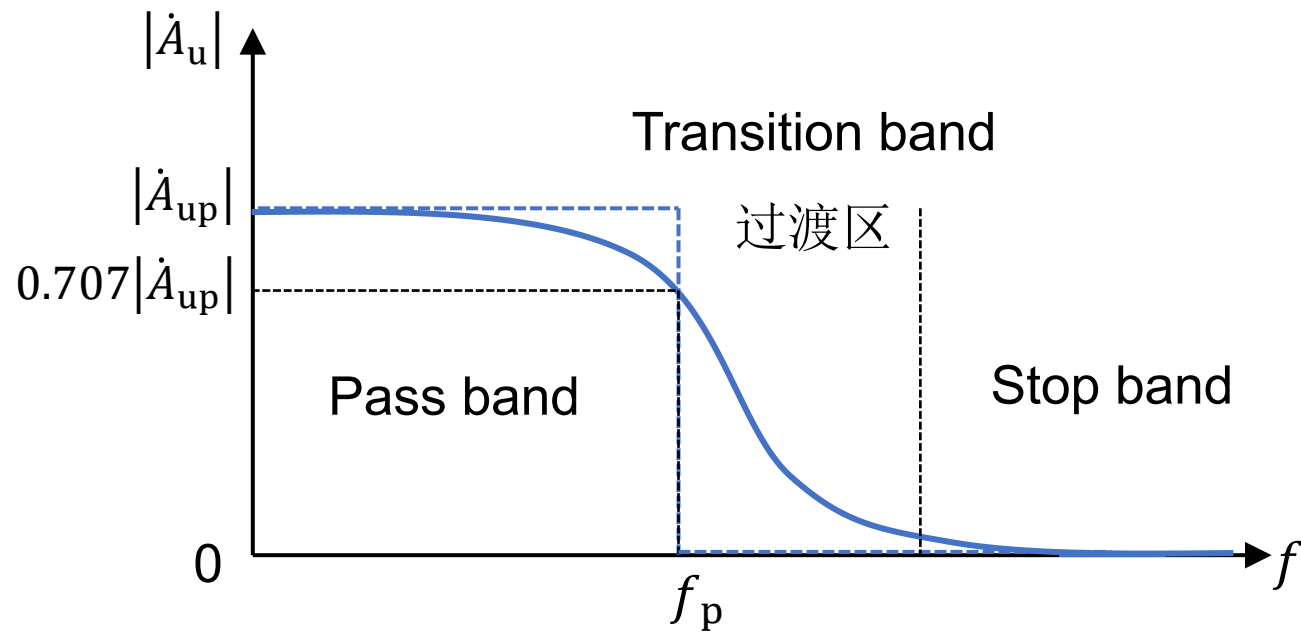


Band elimination filter



Real filters 实际滤波器

Low pass filter



Based on electronic components:

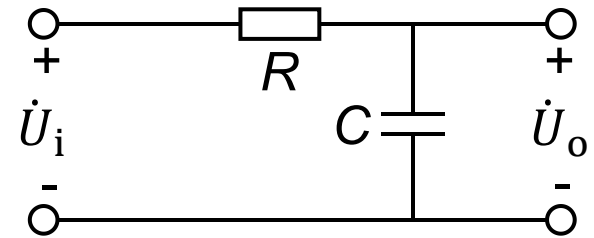
- 
- Passive filter circuits 无源滤波电路
 - Active filter circuits 有源滤波电路

Passive filter circuits 无源滤波电路

Made of resistors, capacitors and inductors

Merits { Simple

High frequency performance is better



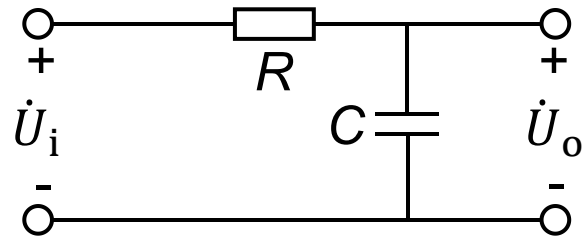
Drawback {

Large load effect: 负载效应大

Space consumption: inductor has large size

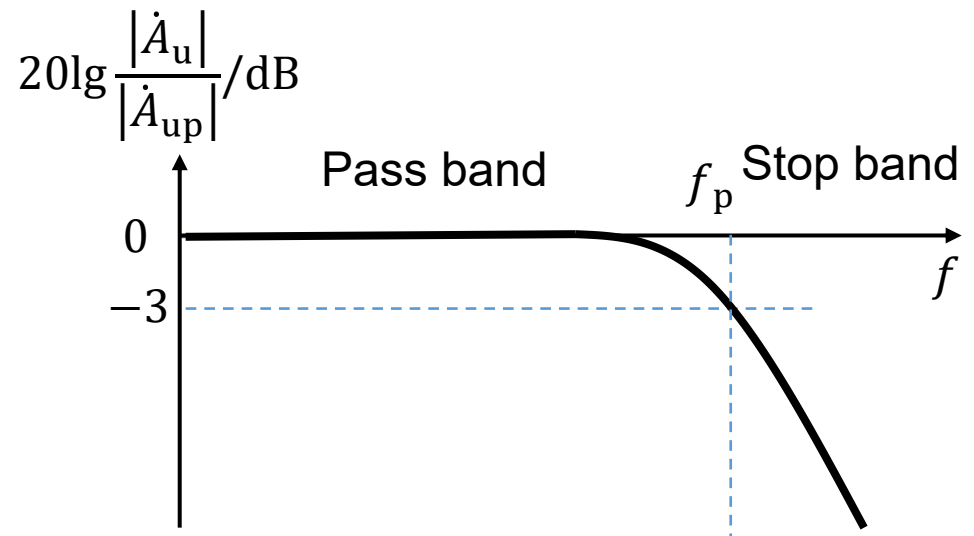
Passive filter circuits 无源滤波电路

$$\dot{A}_u = \frac{\dot{U}_o}{\dot{U}_i} = \frac{\frac{1}{j\omega C}}{\frac{1}{j\omega C} + R}$$



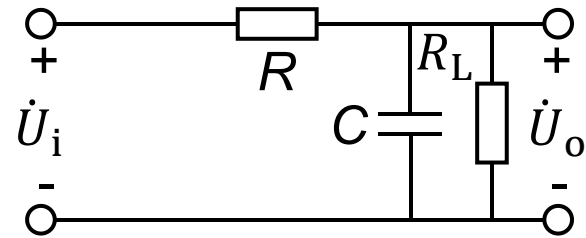
$$|\dot{A}_u| = \frac{|\dot{A}_{up}|}{\sqrt{1 + \frac{f^2}{f_p^2}}}$$

$$\dot{A}_{up} = 1, \quad f_p = \frac{1}{2\pi RC}$$



Passive filter circuits 无源滤波电路

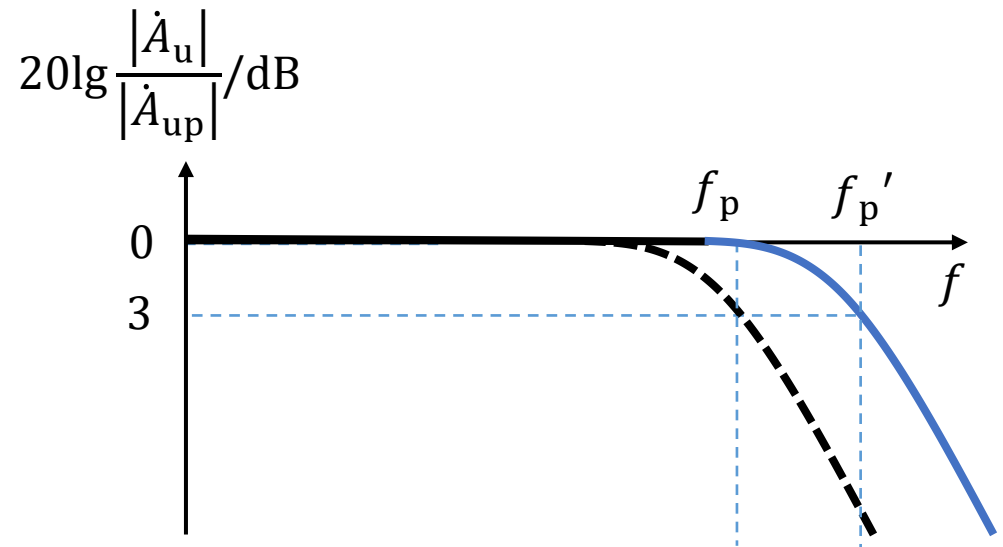
$$\dot{A}_u = \frac{\dot{U}_o}{\dot{U}_i} = \frac{\frac{1}{j\omega C} // R_L}{\frac{1}{j\omega C} // R_L + R}$$



$$|\dot{A}_u| = \frac{|\dot{A}_{up}|}{\sqrt{1 + \frac{f^2}{f_p'^2}}}$$

$$\dot{A}_{up} = \frac{R_L}{R_L + R}$$

$$f_p' = \frac{1}{2\pi(R // R_L)C}$$



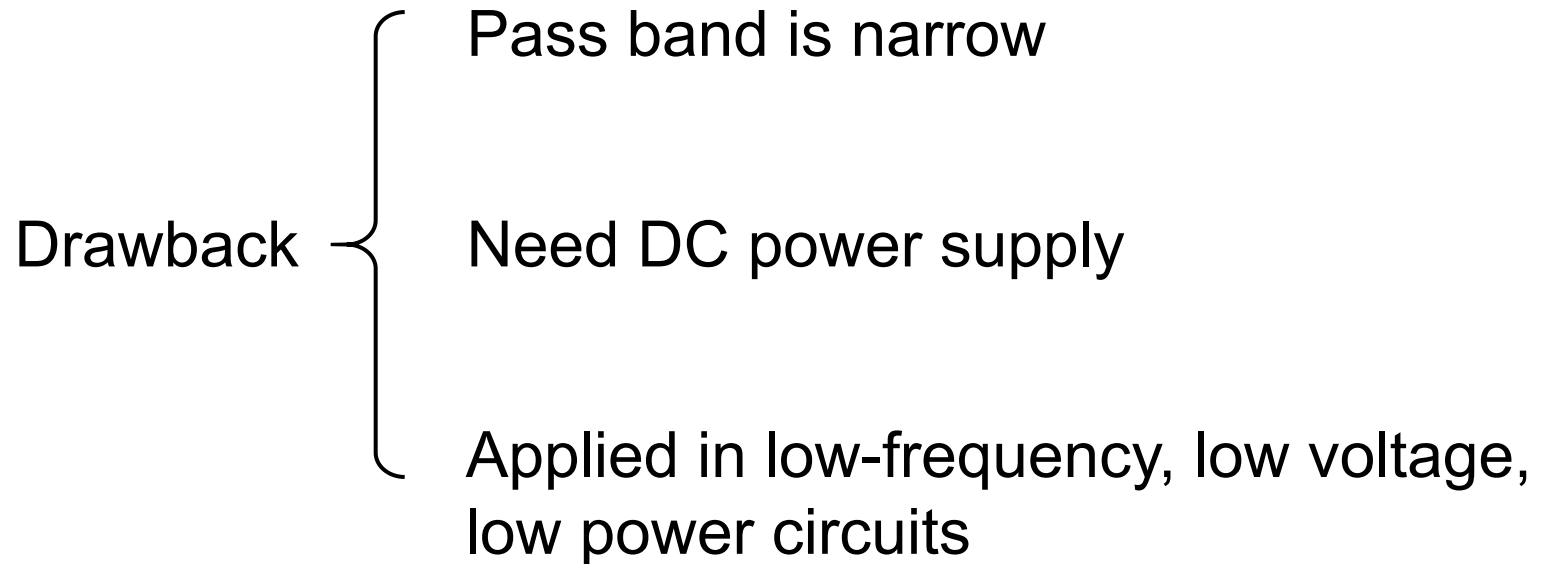
Active filter circuits 有源滤波电路

Made of resistors, capacitors and active devices

- Merits
- Small size
 - Can amplifier signals in pass band
 - Small load effect 负载效应小**
 - Can have multistage to realize high order filter circuits 可以多级相联，组成高阶滤波器

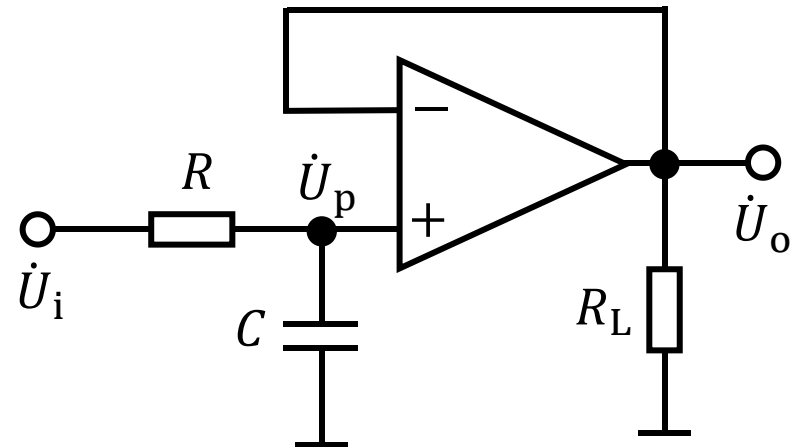
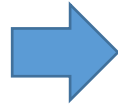
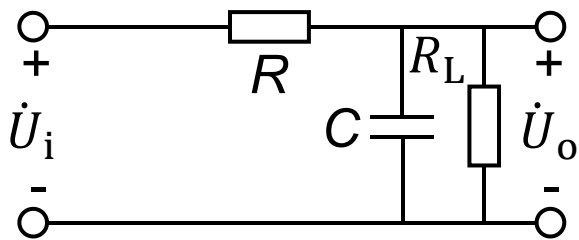
Active filter circuits 有源滤波电路

Made of resistors, capacitors and active devices



Active filter circuits 有源滤波电路

Use voltage follower to separate load resistor with filter circuits



$$\dot{U}_o = \dot{U}_p = \frac{\frac{1}{j\omega C}}{\frac{1}{j\omega C} + R}$$

8.2 Low pass active filters 低通有源滤波器

(1) First-order filters 一阶滤波器

Laplace transform 拉氏变换

Real number Complex number

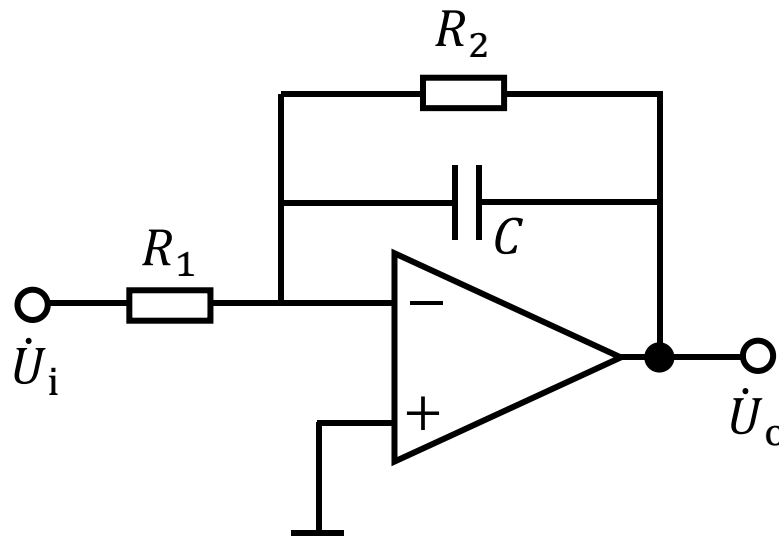
t $\longleftrightarrow$ s

$u_I(t)$ $\longleftrightarrow$ $U_I(s)$

R $\longleftrightarrow$ $Z(s) = R$

Z_C $\longleftrightarrow$ $Z(s) = 1/sC$

When $s = j\omega$: $U_I(s) = \dot{U}_I$, $A_u(s) = \dot{A}_u$



(1) First-order filters 一阶滤波器

Transfer function 传递函数

$$A_u(s) = \frac{U_o(s)}{U_i(s)}$$

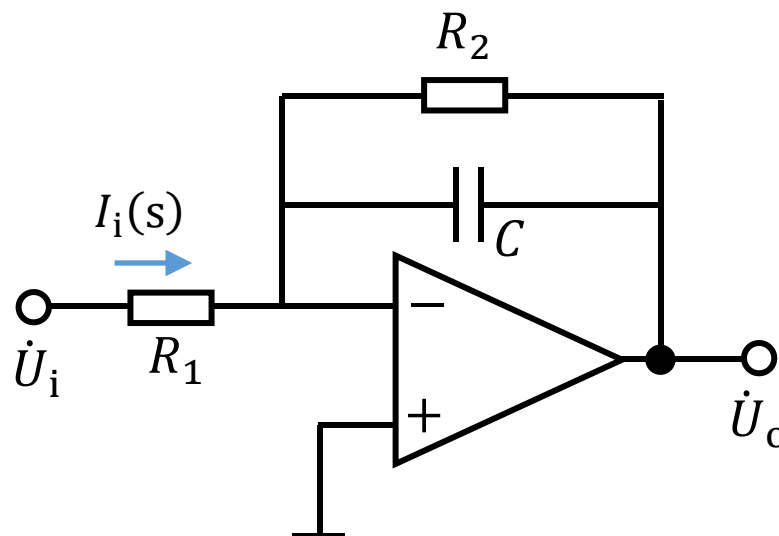
$$U_i(s) = R_1 I_i(s)$$

$$U_o(s)$$

$$= - \left(R_2 // \frac{1}{sC} \right) I_i(s)$$

$$A_u(s) = - \frac{R_2}{R_1} \frac{1}{1 + sR_2C}$$

Inverted input 反相输入



(1) First-order filters 一阶滤波器

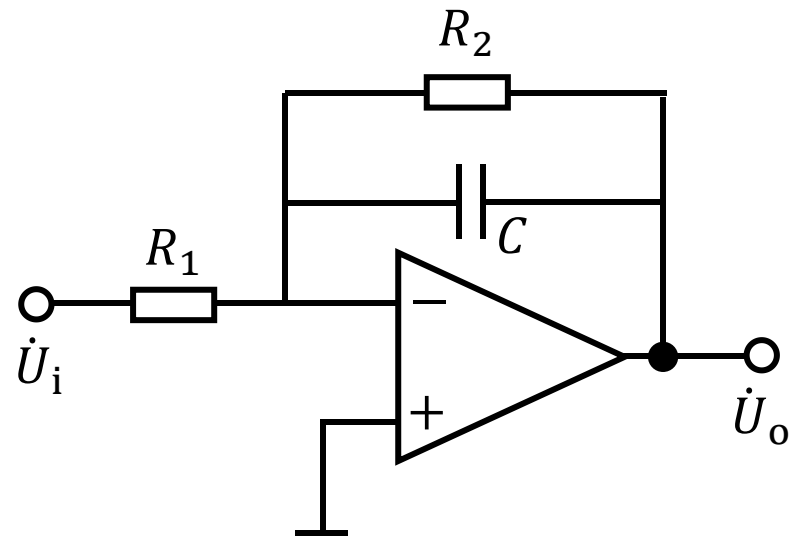
$$A_u(s) = -\frac{R_2}{R_1} \frac{1}{1 + sR_2C}$$

When $s = j\omega$: $A_u(s) = \dot{A}_u$

$$\dot{A}_u = -\frac{R_2}{R_1} \frac{1}{1 + j\omega R_2C}$$

$$\dot{A}_u = \frac{\dot{A}_{up}}{1 + j\frac{f}{f_p}}$$

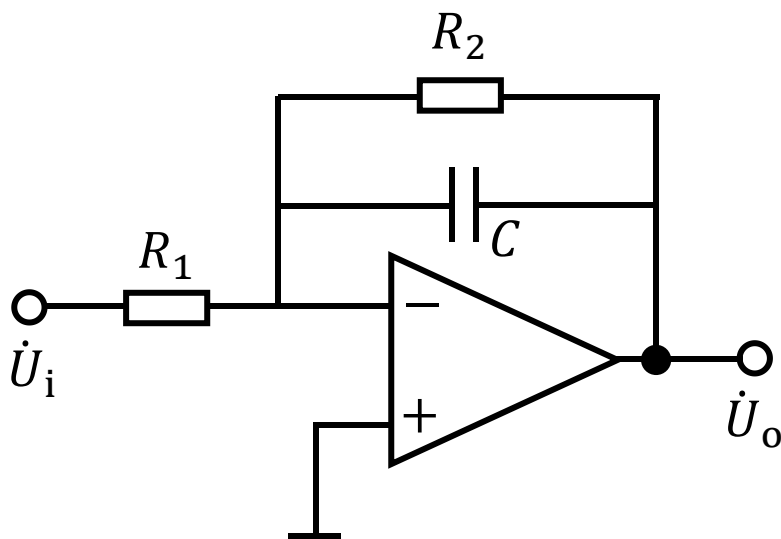
$$\dot{A}_{up} = -\frac{R_2}{R_1} \quad f_p = f_0 = \frac{1}{2\pi R_2C}$$



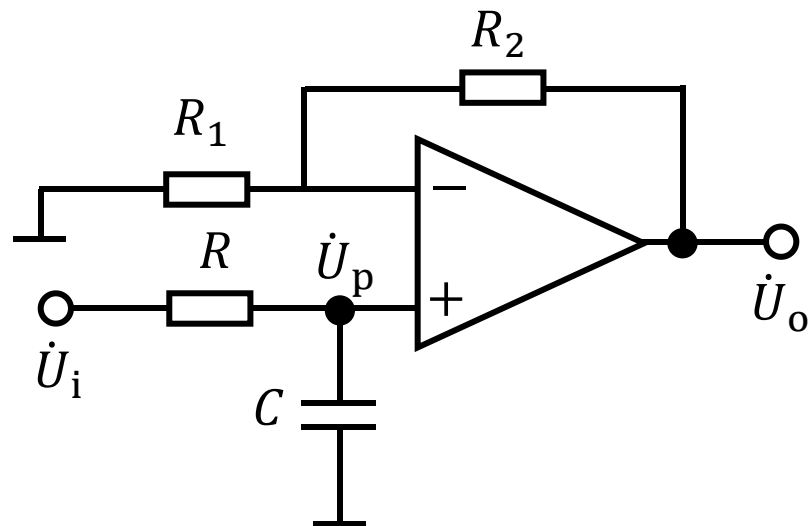
First order: f

(1) First-order filters 一阶滤波器

Inverted input 反相输入

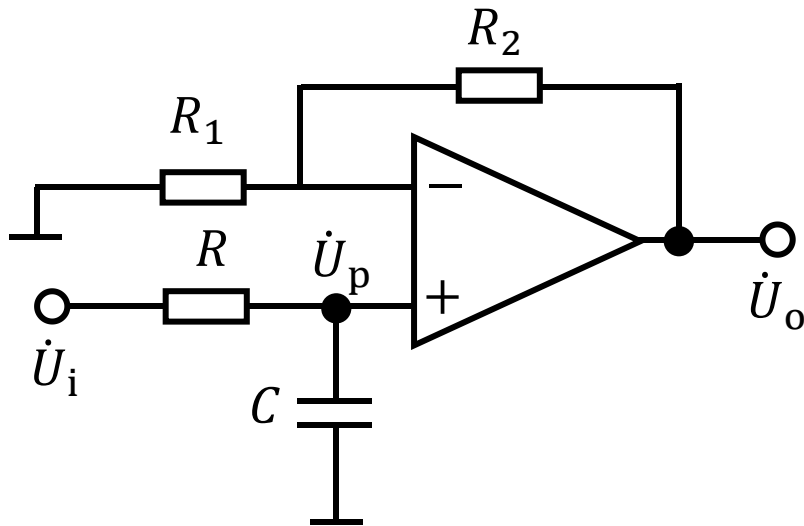


Non-inverted input 同相输入



(1) First-order filters 一阶滤波器

Non-inverted input 同相输入



$$\dot{A}_u = \frac{\dot{A}_{up}}{1 + j \frac{f}{f_p}}$$

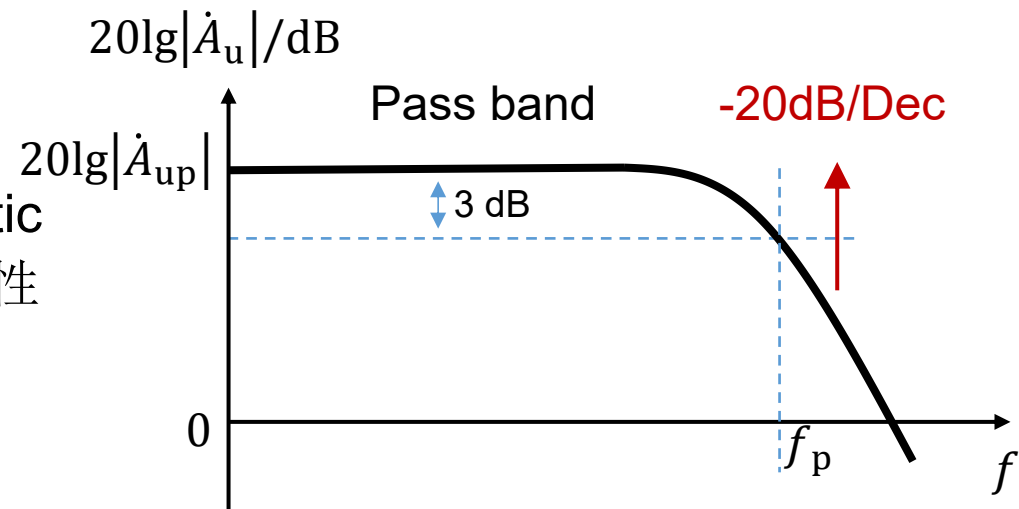
$$\dot{A}_{up} = 1 + \frac{R_2}{R_1}, \quad f_p = \frac{1}{2\pi RC}$$

(1) First-order filters 一阶滤波器

$$\dot{A}_u = \frac{\dot{A}_{up}}{1 + j \frac{f}{f_p}}$$

Amplitude frequency characteristic
幅频特性

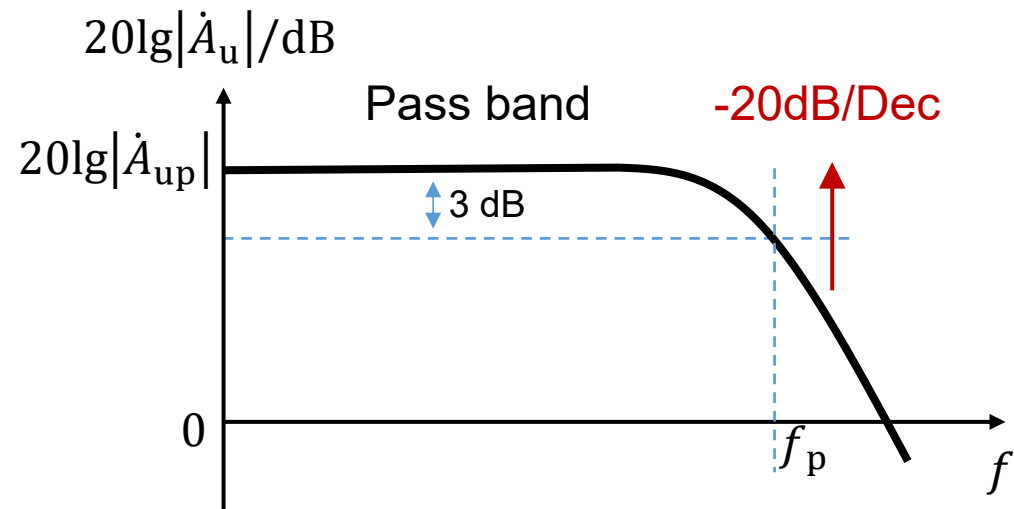
$$|\dot{A}_u| = \frac{|\dot{A}_{up}|}{\sqrt{1 + \frac{f^2}{f_p^2}}}$$



(1) First-order filters 一阶滤波器

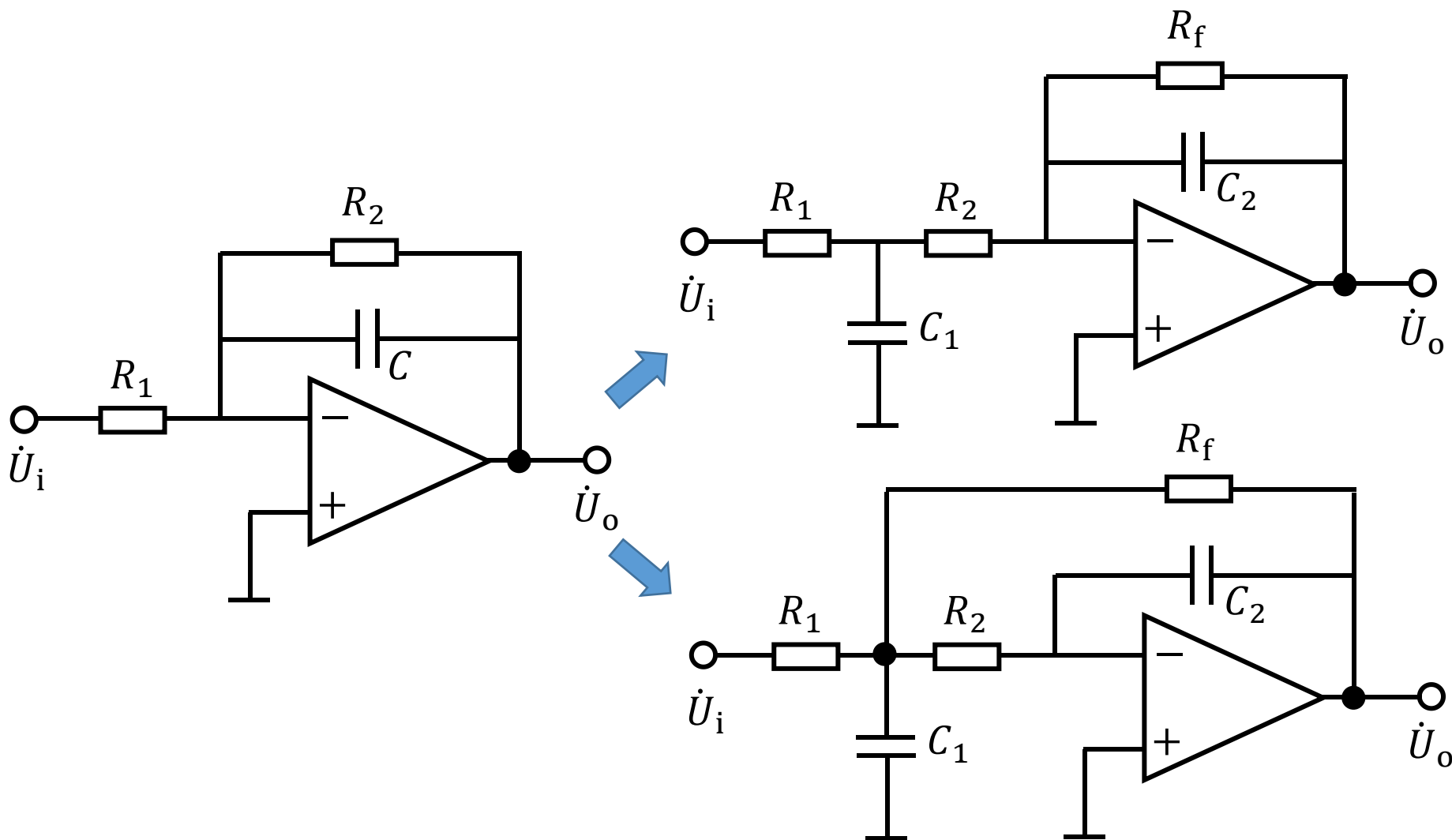
Merit: circuit is simple

Drawback: slow signal
attenuation: 20 dB/Dec
信号衰减慢

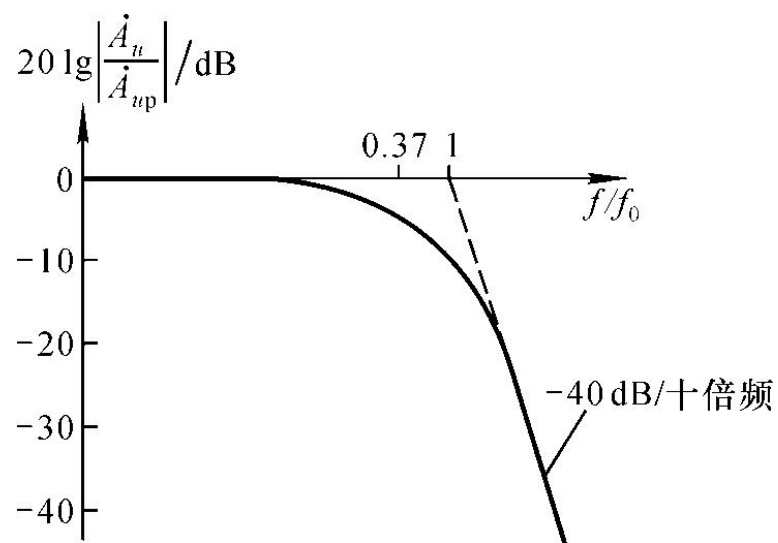
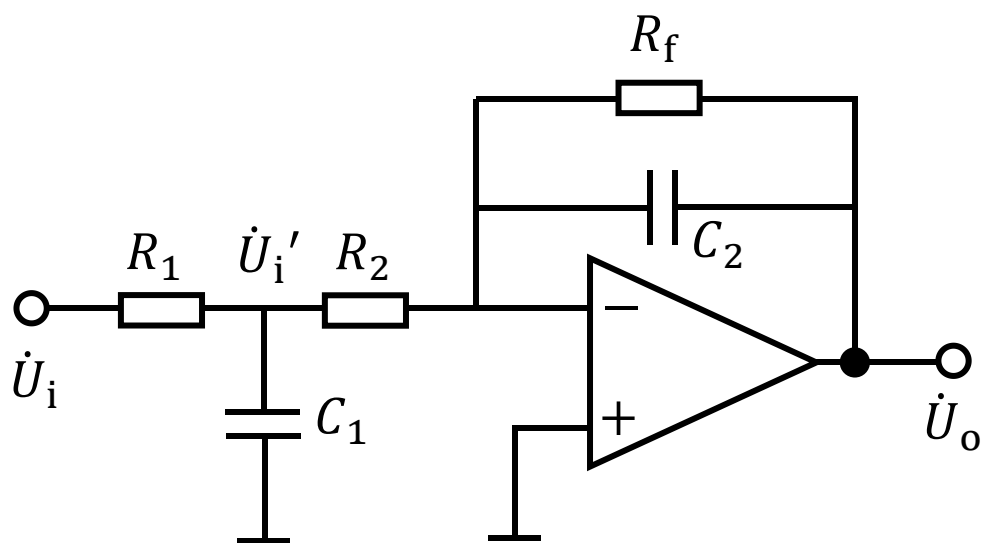


(2) Second-order filters 二阶滤波器

Signal attenuation rate -20 dB/Dec $\rightarrow$ -40 dB/Dec



(2) Second-order filters 二阶滤波器



(2) Second-order filters 二阶滤波器

Transfer function 传递函数

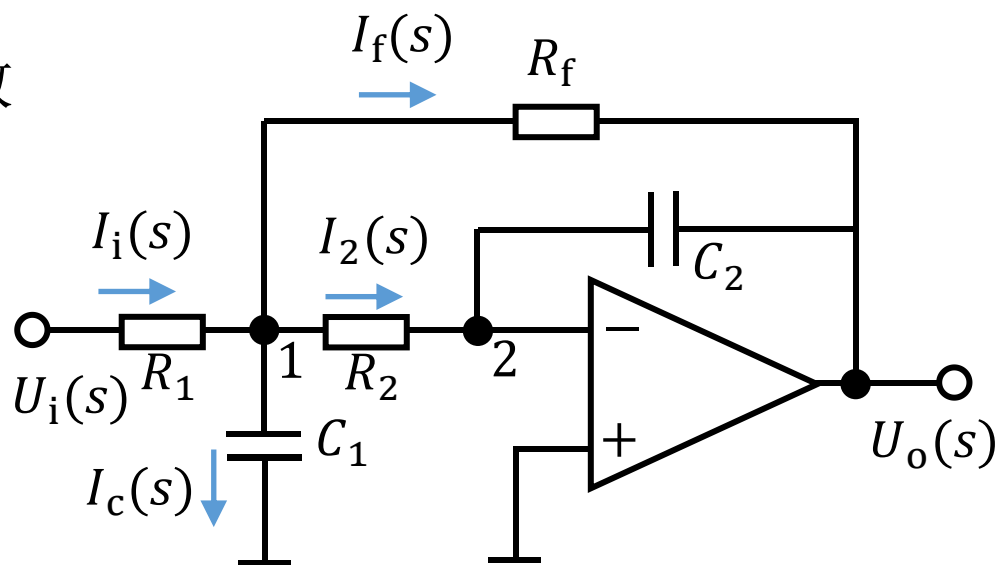
Node 1 节点1:

$$I_i(s) = I_c(s) + I_f(s) + I_2(s)$$

$$\frac{U_i(s) - U_1(s)}{R_1} = \frac{U_1(s)}{1/sC_1} + \frac{U_1(s) - U_o(s)}{R_f} + \frac{U_1(s)}{R_2}$$

Node 2 节点2:

$$\frac{U_1(s)}{R_2} = \frac{-U_o(s)}{1/sC_2}$$



(2) Second-order filters 二阶滤波器

Transfer function 传递函数

$$A_u(s) = \frac{U_o(s)}{U_i(s)} = -\frac{R_f}{R_1} \frac{1}{1 + sC_2R_2R_f\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_f}\right) + s^2C_1C_2R_2R_f}$$

$$\left\{ \begin{array}{ll} \omega_0 = 2\pi f_0 & \text{Characteristic angular frequency} \\ = 1/\sqrt{C_1C_2R_2R_f} & \text{特征角频率} \\ Q = (R_1 // R_2 // R_f) \sqrt{\frac{C_1}{C_2R_2R_f}} & \text{Quality factor} \\ & \text{品质因子} \end{array} \right.$$

$$A_u(s) = -\frac{R_f}{R_1} \frac{\omega_0^2}{s^2 + s\omega_0/Q + \omega_0^2}$$

(2) Second-order filters 二阶滤波器

Frequency response $s = j\omega$

$$\dot{A}_u = A_u(j\omega) = \dot{A}_{up} \frac{1}{1 - \frac{\omega^2}{\omega_0^2} + j \frac{\omega}{\omega_0 Q}}, \quad \dot{A}_{up} = -\frac{R_f}{R_1}$$

Amplitude frequency characteristic

$$|\dot{A}_u| = \frac{|\dot{A}_{up}|}{\sqrt{\left(1 - \frac{\omega^2}{\omega_0^2}\right)^2 + \frac{\omega^2}{\omega_0^2 Q^2}}}$$

(2) Second-order filters 二阶滤波器

Amplitude frequency characteristic

$$|\dot{A}_u| = \frac{|\dot{A}_{up}|}{\sqrt{\left(1 - \frac{f^2}{f_0^2}\right)^2 + \frac{f^2}{f_0^2 Q^2}}} \quad \frac{\omega}{\omega_0} = \frac{f}{f_0}$$

$$(1) f \ll f_0, \quad |\dot{A}_u| \rightarrow |\dot{A}_{up}|$$

$$(2) f \gg f_0, \quad |\dot{A}_u| \rightarrow 0$$

Low pass filter!

(2) Second-order filters 二阶滤波器

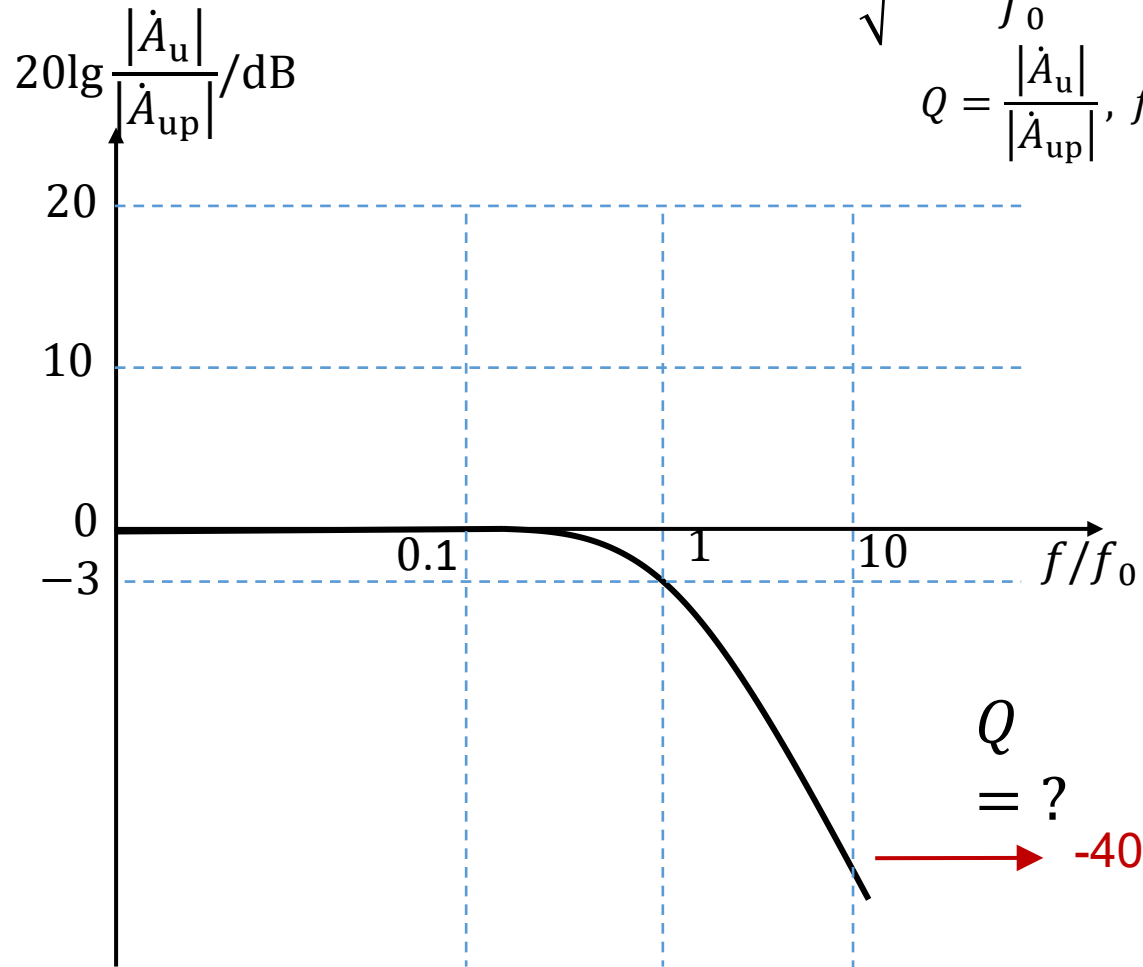
Amplitude frequency characteristic

$$|\dot{A}_u| = \frac{|\dot{A}_{up}|}{\sqrt{\left(1 - \frac{f^2}{f_0^2}\right)^2 + \frac{f^2}{f_0^2 Q^2}}}$$

$$(3) f = f_0, \begin{cases} |\dot{A}_u| = Q|\dot{A}_{up}| \\ Q = |\dot{A}_u|/|\dot{A}_{up}| \end{cases}$$

(2) Second-order filters 二阶滤波器

Amplitude frequency characteristic



$$|\dot{A}_u| = \frac{|\dot{A}_{up}|}{\sqrt{(1 - \frac{f^2}{f_0^2})^2 + \frac{f^2}{f_0^2} Q^2}}$$
$$Q = \frac{|\dot{A}_u|}{|\dot{A}_{up}|}, f = f_0$$

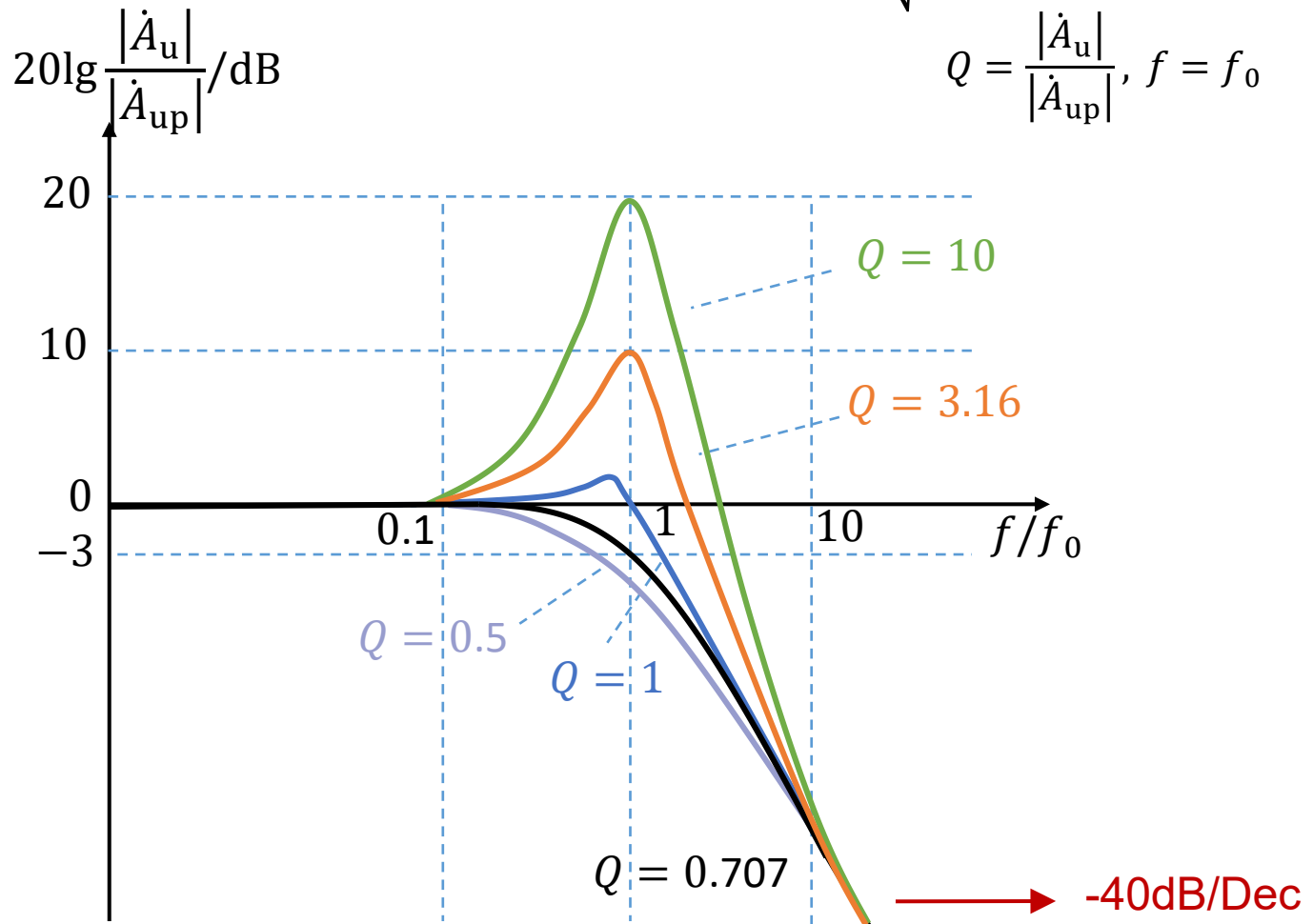
Q
 $= ?$

-40dB/Dec

(2) Second-order filters 二阶滤波器

Amplitude frequency characteristic

$$|\dot{A}_u| = \frac{|\dot{A}_{up}|}{\sqrt{(1 - \frac{f^2}{f_0^2})^2 + \frac{f^2}{f_0^2 Q^2}}}$$
$$Q = \frac{|\dot{A}_u|}{|\dot{A}_{up}|}, f = f_0$$



(2) Second-order filters 二阶滤波器

$$|\dot{A}_u| = \frac{|\dot{A}_{up}|}{\sqrt{(1 - \frac{f^2}{f_0^2})^2 + \frac{f^2}{f_0^2 Q^2}}}$$

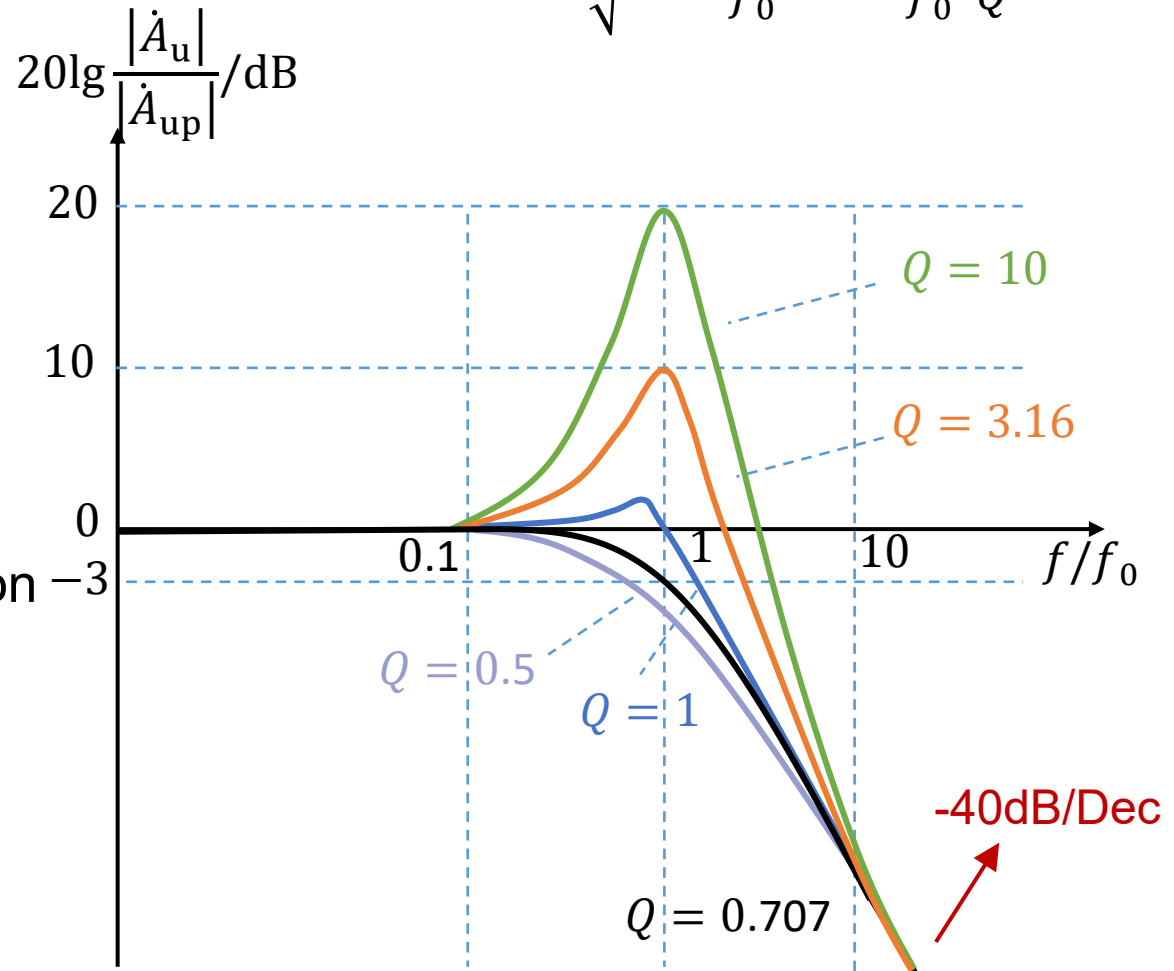
(a) $Q > 1/\sqrt{2}$

There is a peak

Larger Q , higher peak

$Q \rightarrow \infty$, self-excited oscillation

Chebyshev filters
切比雪夫滤波器



(2) Second-order filters 二阶滤波器

$$|\dot{A}_u| = \frac{|\dot{A}_{up}|}{\sqrt{(1 - \frac{f^2}{f_0^2})^2 + \frac{f^2}{f_0^2 Q^2}}}$$

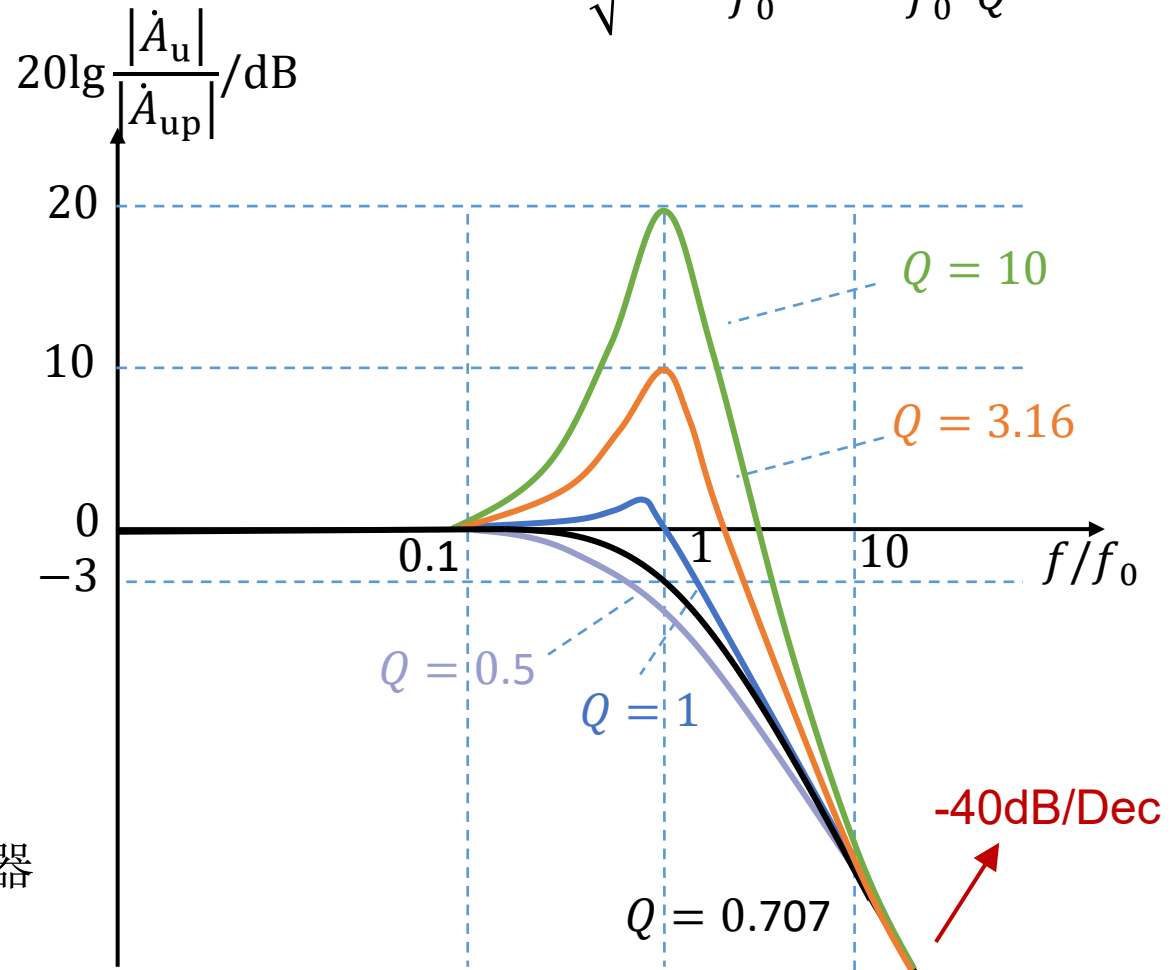
(b) $Q < 1/\sqrt{2}$

No peak

Signal attenuation in pass band is relatively large
通带信号衰减较大

Smaller Q, higher signal attenuation

Bessel filter 贝塞尔滤波器



(2) Second-order filters 二阶滤波器

$$|\dot{A}_u| = \frac{|\dot{A}_{up}|}{\sqrt{(1 - \frac{f^2}{f_0^2})^2 + \frac{f^2}{f_0^2 Q^2}}}$$

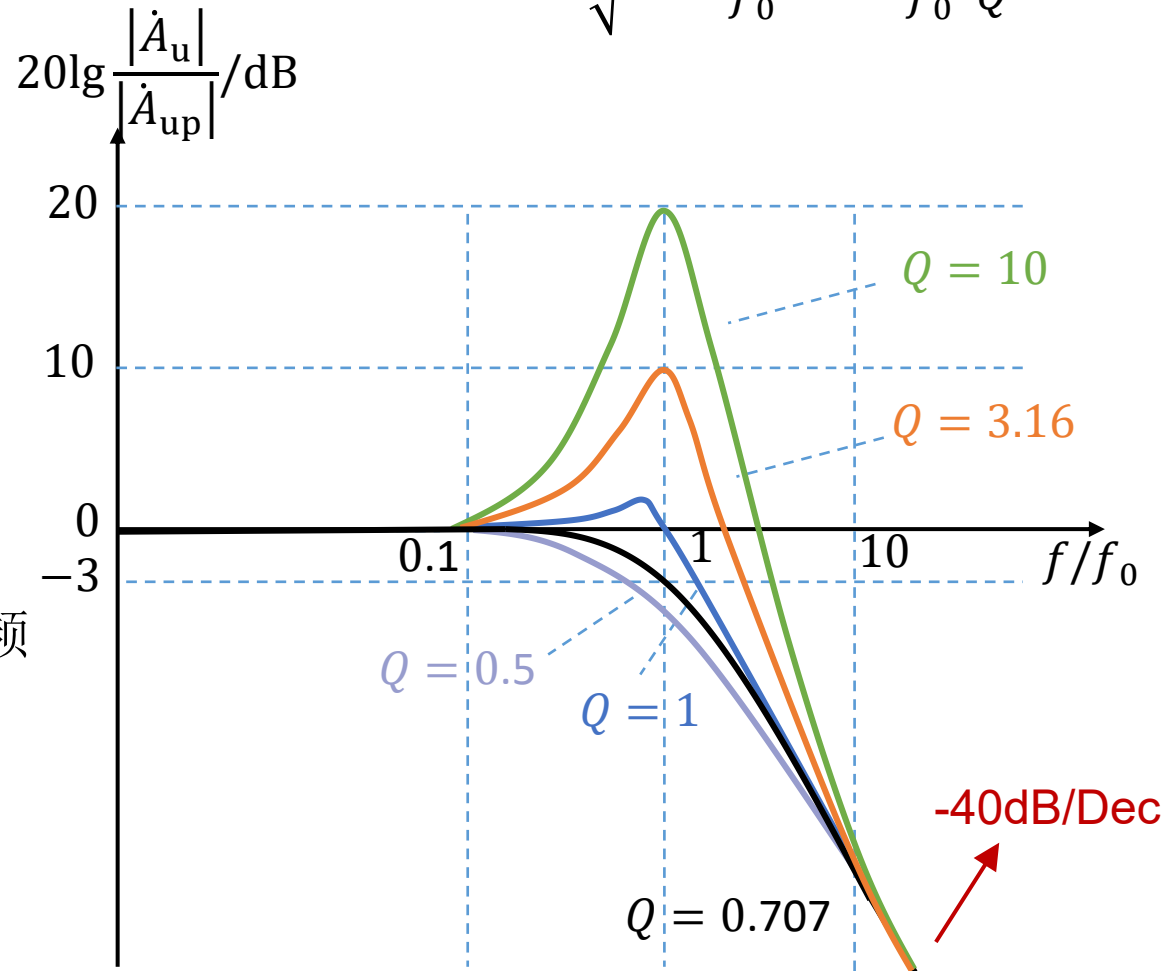
(c) $Q = 1/\sqrt{2}$

No peak

Signal attenuation in pass band is relatively large
通带信号衰减较大

Amplitude-frequency characteristic is flattest
幅频特性最平坦

Maximum flat filter
/Butterworth filter 最大平坦/巴特沃斯滤波器



(2) Second-order filters 二阶滤波器

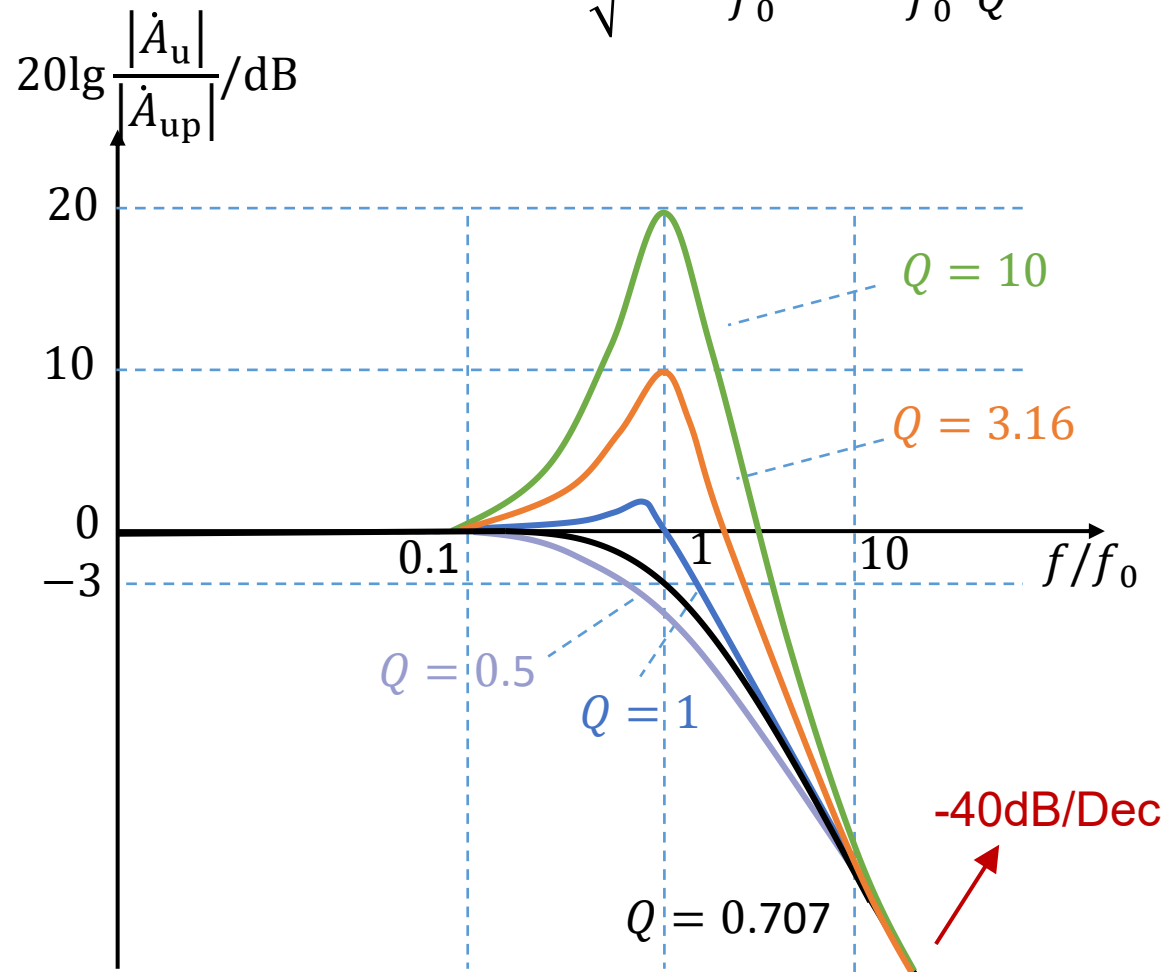
$$|\dot{A}_u| = \frac{|\dot{A}_{up}|}{\sqrt{(1 - \frac{f^2}{f_0^2})^2 + \frac{f^2}{f_0^2 Q^2}}}$$

(c) $Q = 1/\sqrt{2}$

$$|\dot{A}_u| = \frac{|\dot{A}_{up}|}{\sqrt{1 + \frac{f^4}{f_0^4}}}$$

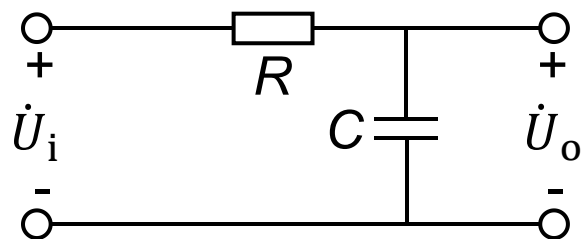
When $f = f_0$

$$|\dot{A}_u| = \frac{|\dot{A}_{up}|}{\sqrt{2}}$$

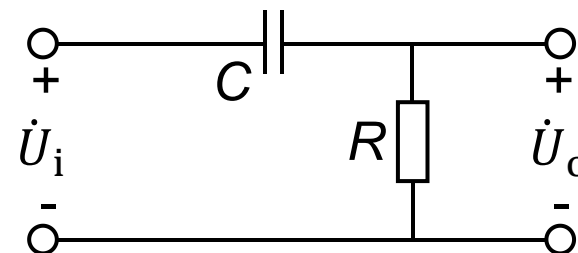


8.2 High pass active filters 高通有源滤波器

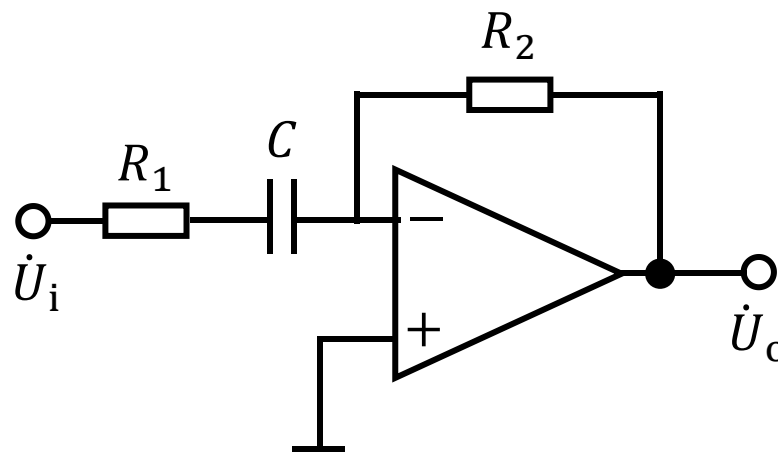
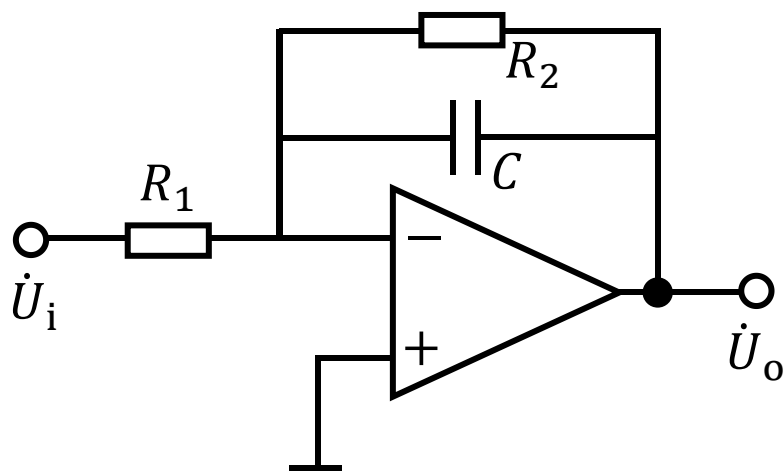
(1) First-order filters 一阶滤波器



Low pass



High pass



(1) First-order filters 一阶滤波器

Transfer function 传递函数

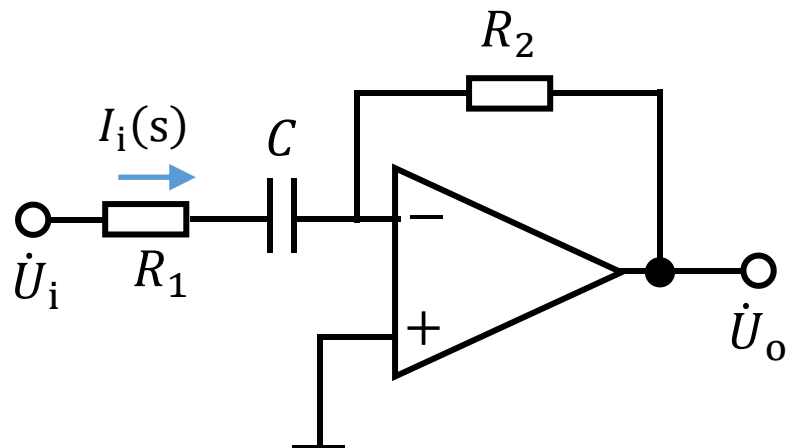
$$A_u(s) = \frac{U_o(s)}{U_i(s)}$$

$$U_i(s) = (R_1 + \frac{1}{sC})I_i(s)$$

$$U_o(s) = -R_2 I_i(s)$$

$$A_u(s) = -\frac{R_2}{R_1} \frac{1}{1 + 1/sR_1C}$$

Inverted input 反相输入



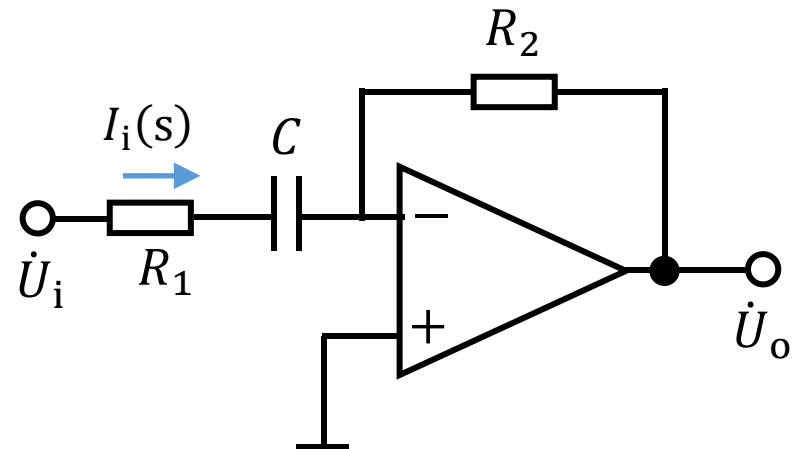
(1) First-order filters 一阶滤波器

$$A_u(s) = -\frac{R_2}{R_1} \frac{1}{1 + 1/sR_1C}$$

When $s = j\omega$: $A_u(s) = \dot{A}_u$

$$\dot{A}_u = \frac{\dot{A}_{up}}{1 + \frac{f_p}{jf}}$$

$$\dot{A}_{up} = -\frac{R_2}{R_1} \quad f_p = f_0 = \frac{1}{2\pi R_1 C}$$



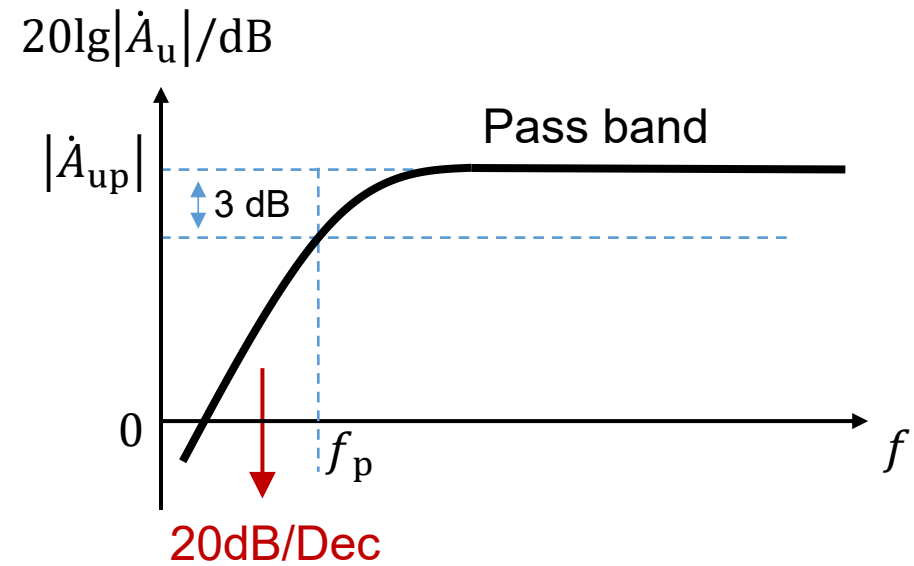
First order: f

(1) First-order filters 一阶滤波器

$$\dot{A}_u = \frac{\dot{A}_{up}}{1 - j \frac{f_p}{f}}$$

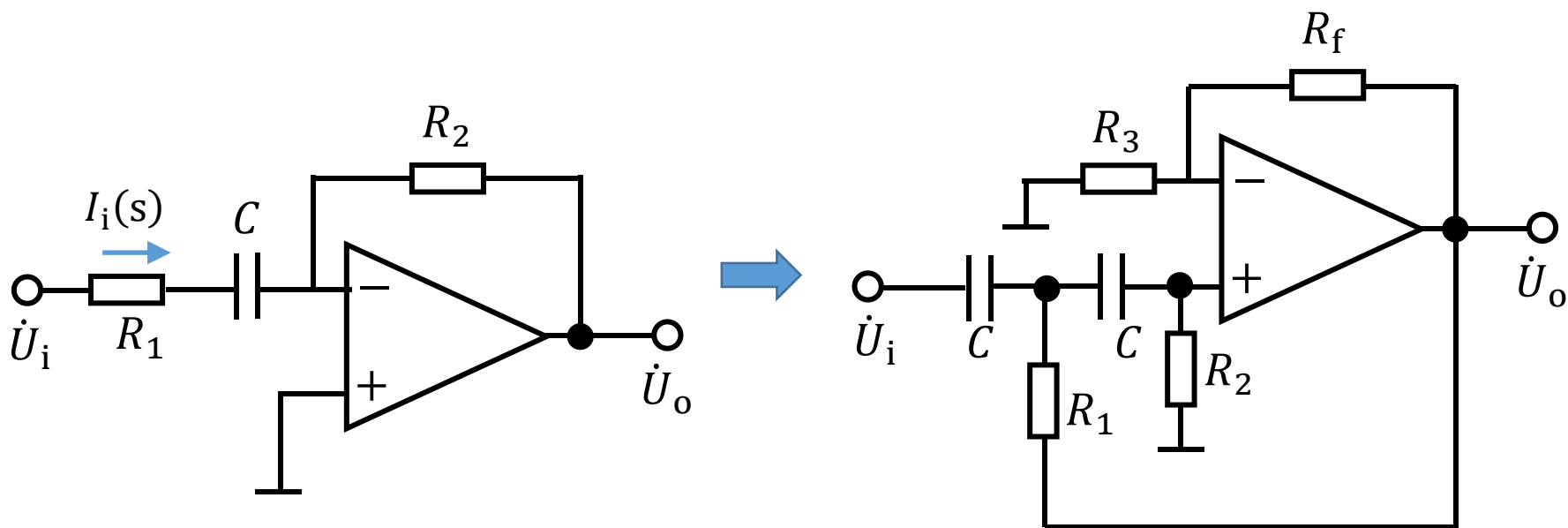
Amplitude frequency characteristic
幅频特性

$$|\dot{A}_u| = \frac{|\dot{A}_{up}|}{\sqrt{1 + \frac{f_p^2}{f^2}}}$$



(2) Second-order filters 二阶滤波器

Signal attenuation rate 20 dB/Dec $\rightarrow$ 40 dB/Dec



(2) Second-order filters 二阶滤波器

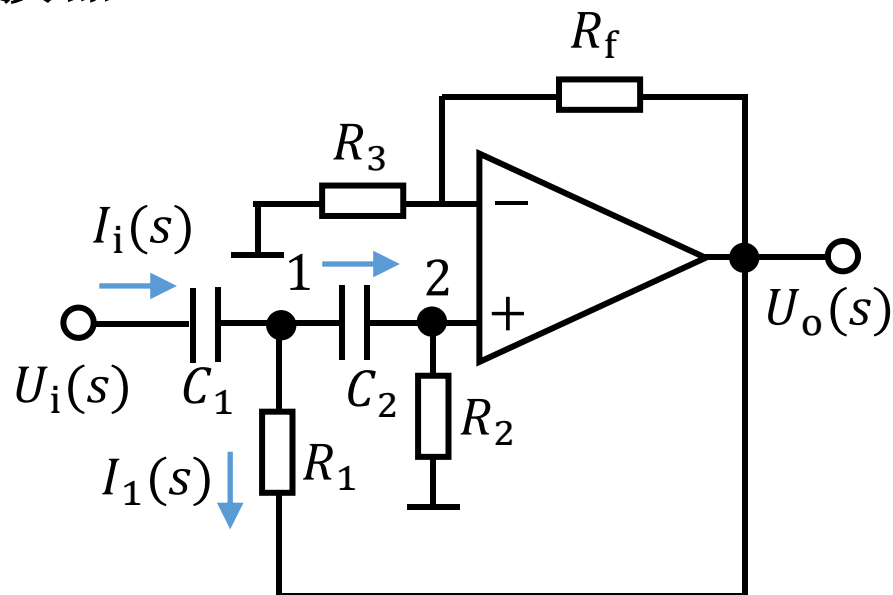
Transfer function 传递函数?

$$U_o(s) = (1 + \frac{R_f}{R_3})U_2(s)$$

Node 1 节点1:

$$\frac{U_i(s) - U_1(s)}{1/sC_1} = \frac{U_1(s) - U_2(s)}{1/sC_2} + \frac{U_1(s) - U_o(s)}{R_1}$$

Node 2 节点2:
$$\frac{U_1(s) - U_2(s)}{1/sC_2} = \frac{U_2(s)}{R_2}$$



(2) Second-order filters 二阶滤波器

Transfer function 传递函数

$$A_u(s) = \frac{U_o(s)}{U_i(s)}$$

$$= A_{up} \frac{s^2}{s^2 + s \frac{C_1 R_1 + C_2 R_1 + C_2 R_2 (1 - A_{up})}{C_1 C_2 R_1 R_2} + \frac{1}{C_1 C_2 R_1 R_2}}, \quad A_{up} = 1 + \frac{R_f}{R_3}$$

$$\omega_0 = 2\pi f_0? \quad Q?$$

(2) Second-order filters 二阶滤波器

Transfer function 传递函数

$$A_u(s) = A_{up} \frac{s^2}{s^2 + s \frac{C_1 R_1 + C_2 R_1 + C_2 R_2 (1 - A_{up})}{C_1 C_2 R_1 R_2} + \frac{1}{C_1 C_2 R_1 R_2}}$$
$$, A_{up} = 1 + \frac{R_f}{R_3}$$

$$A_u(s) = A_{up} \frac{s^2}{s^2 + s\omega_0/Q + \omega_0^2}$$

$$\left\{ \begin{array}{l} \omega_0 = 2\pi f_0 \\ \quad = 1/\sqrt{C_1 C_2 R_1 R_2} \\ Q = \frac{\sqrt{C_1 C_2 R_1 R_2}}{C_1 R_1 + C_2 R_1 + C_2 R_2 (1 - A_{up})} \end{array} \right. \quad \begin{array}{l} \text{特征角频率} \\ \\ \text{Quality factor} \\ \text{品质因子} \end{array}$$

(2) Second-order filters 二阶滤波器

Frequency response

$$s = j\omega$$

$$A_u(s) = A_{up} \frac{s^2}{s^2 + s\omega_0/Q + \omega_0^2}$$

$$\dot{A}_u = A_u(j\omega) = -\dot{A}_{up} \frac{1}{\frac{\omega_0^2}{\omega^2} - 1 + j\frac{\omega_0}{\omega Q}}, \quad \dot{A}_{up} = 1 + \frac{R_f}{R_3}$$

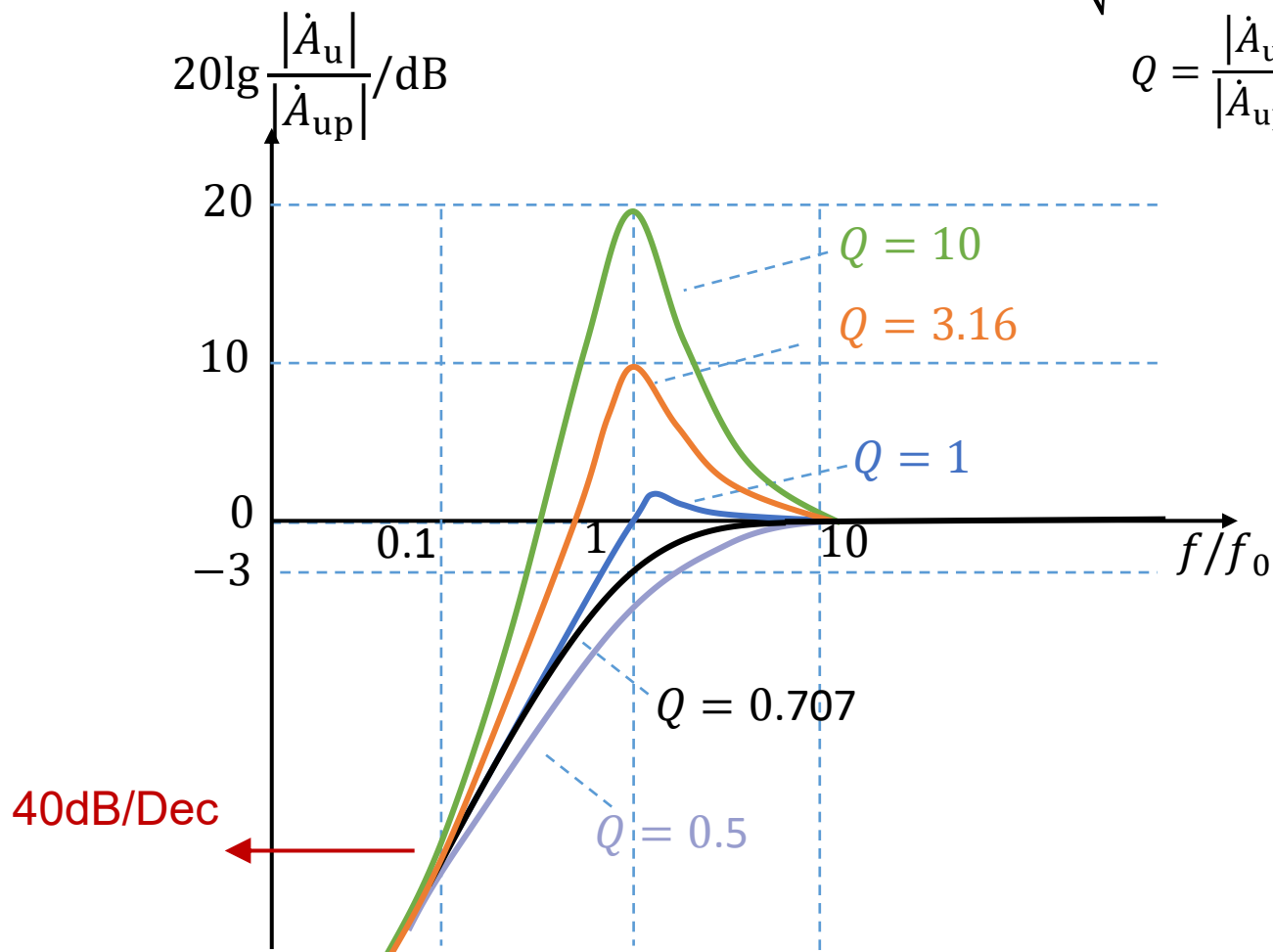
Amplitude frequency characteristic

$$|\dot{A}_u| = \frac{|\dot{A}_{up}|}{\sqrt{\left(\frac{\omega_0^2}{\omega^2} - 1\right)^2 + \frac{\omega_0^2}{\omega^2 Q^2}}} = \frac{|\dot{A}_{up}|}{\sqrt{\left(\frac{f_0^2}{f^2} - 1\right)^2 + \frac{f_0^2}{f^2 Q^2}}}$$

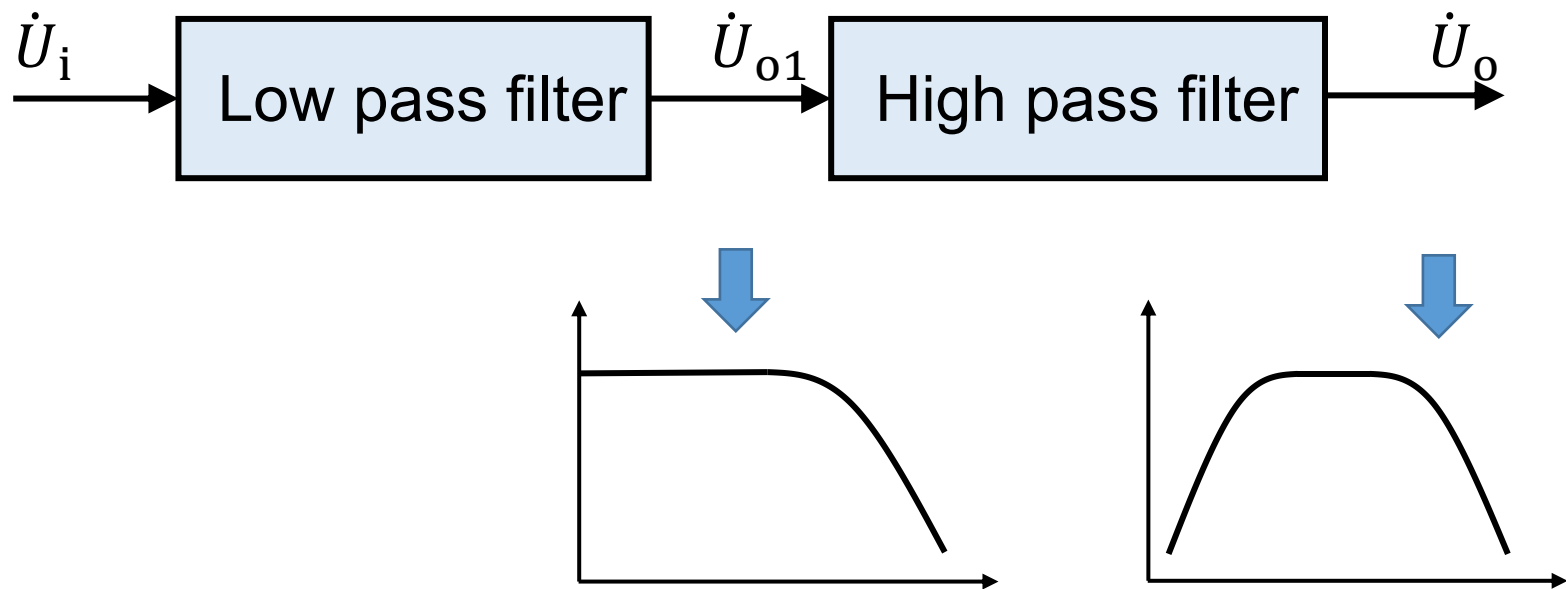
(2) Second-order filters 二阶滤波器

Amplitude frequency characteristic

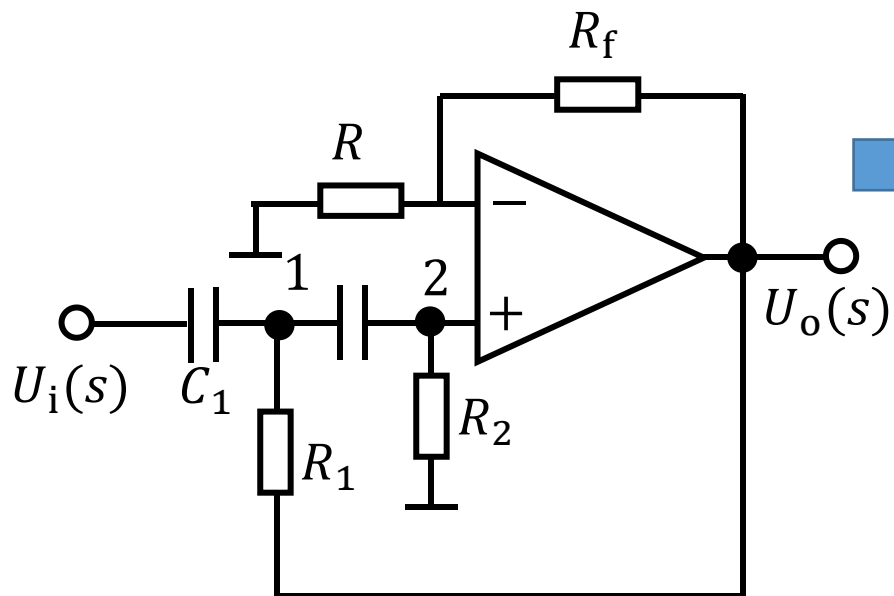
$$|\dot{A}_u| = \frac{|\dot{A}_{up}|}{\sqrt{(\frac{f_0^2}{f^2} - 1)^2 + \frac{f_0^2}{f^2 Q^2}}}$$
$$Q = \frac{|\dot{A}_u|}{|\dot{A}_{up}|}, f = f_0$$



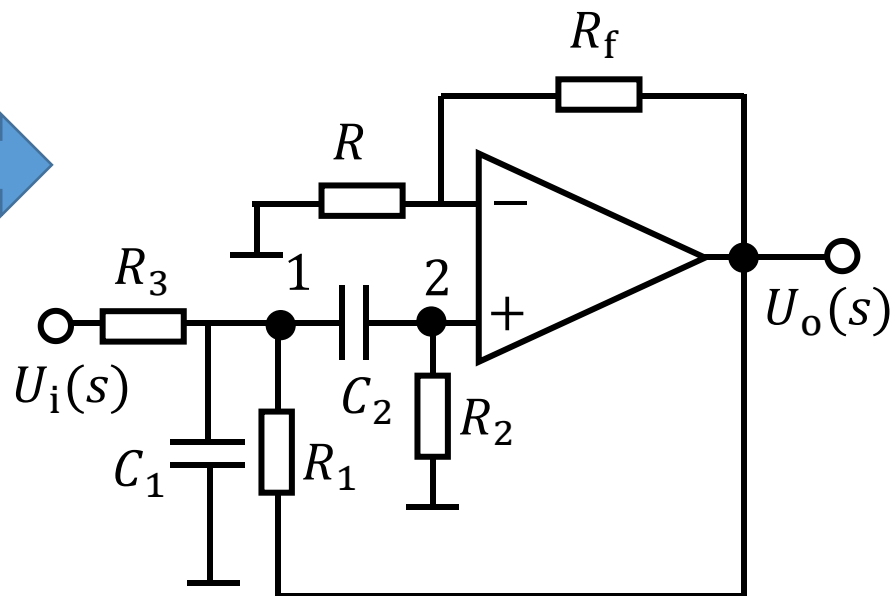
8.3 Band pass active filters 帶通有源濾波器



High Pass filter

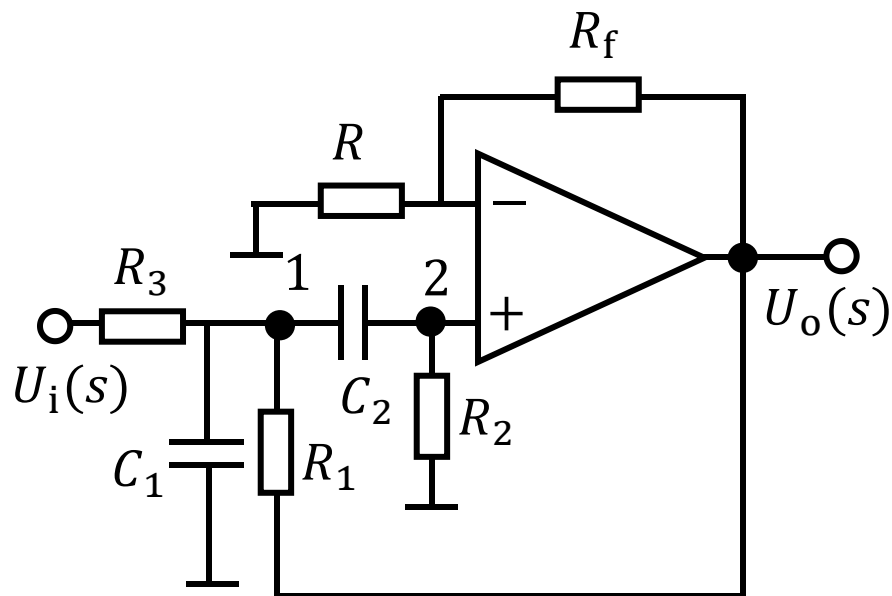


Band Pass filter



Transfer function 传递函数?

$$U_o(s) = \left(1 + \frac{R_f}{R}\right) U_2(s)$$



Node 1 节点1:

$$\frac{U_i(s) - U_1(s)}{R_3} = \frac{U_1(s)}{1/sC_1} + \frac{U_1(s) - U_2(s)}{1/sC_2} + \frac{U_1(s) - U_o(s)}{R_1}$$

Node 2 节点2:

$$\frac{U_1(s) - U_2(s)}{1/sC_2} = \frac{U_2(s)}{R_2}$$

Transfer function 传递函数

$$A_u(s) = \frac{U_o(s)}{U_i(s)}$$

$$= \frac{A_0 s / R_3 C_1}{s^2 + s\omega_0 / Q + \omega_0^2}, \quad A_0 = 1 + \frac{R_f}{R}$$

$$\left\{ \begin{array}{l} \omega_0 = 2\pi f_0 = \sqrt{\frac{R_1 + R_3}{R_1 R_2 R_3 C_1 C_2}} \\ Q = \frac{\omega_0}{1/R_3 C_1 + 1/R_2 C_1 + 1/R_3 C_2 + (1 - A_0)/R_1 C_1} \end{array} \right.$$

Frequency response: $s = j\omega$

$$\dot{A}_u = A_u(j\omega) = -\frac{jA_0\omega/R_3C_1}{\omega_0^2 - \omega^2 + j\omega\omega_0/Q}, \quad \dot{A}_0 = 1 + \frac{R_f}{R}$$

Amplitude frequency characteristic

$$|\dot{A}_u| = \frac{\dot{A}_0/(R_3C_1\omega_0)}{\sqrt{(\frac{\omega_0}{\omega} - \frac{\omega}{\omega_0})^2 + \frac{1}{Q^2}}} = \frac{\dot{A}_0/(R_3C_1\omega_0)}{\sqrt{(\frac{f_0}{f} - \frac{f}{f_0})^2 + \frac{1}{Q^2}}}$$

$$|\dot{A}_u| = \frac{\dot{A}_0 / (R_3 C_1 \omega_0)}{\sqrt{\left(\frac{f_0}{f} - \frac{f}{f_0}\right)^2 + \frac{1}{Q^2}}}$$

(a) When $f = 0$, $|\dot{A}_u| = 0$

(b) When $f \rightarrow \infty$, $|\dot{A}_u| \rightarrow 0$

(c) When $f = f_0$, $|\dot{A}_u| = \frac{\dot{A}_0 Q}{R_3 C_1 \omega_0}$

Band pass filter!

Cut-off frequency

$$\frac{\dot{A}_0/(R_3 C_1 \omega_0)}{\sqrt{(\frac{f_0}{f} - \frac{f}{f_0})^2 + \frac{1}{Q^2}}} = \frac{\dot{A}_0 Q/(R_3 C_1 \omega_0)}{\sqrt{2}}$$



$$(\frac{f_0}{f} - \frac{f}{f_0})^2 = \frac{1}{Q^2}$$



$$\left\{ \begin{array}{l} f_{\text{pH}} = (\sqrt{1 + \frac{1}{4Q^2}} + \frac{1}{2Q})f_0 \\ f_{\text{pL}} = (\sqrt{1 + \frac{1}{4Q^2}} - \frac{1}{2Q})f_0 \end{array} \right.$$

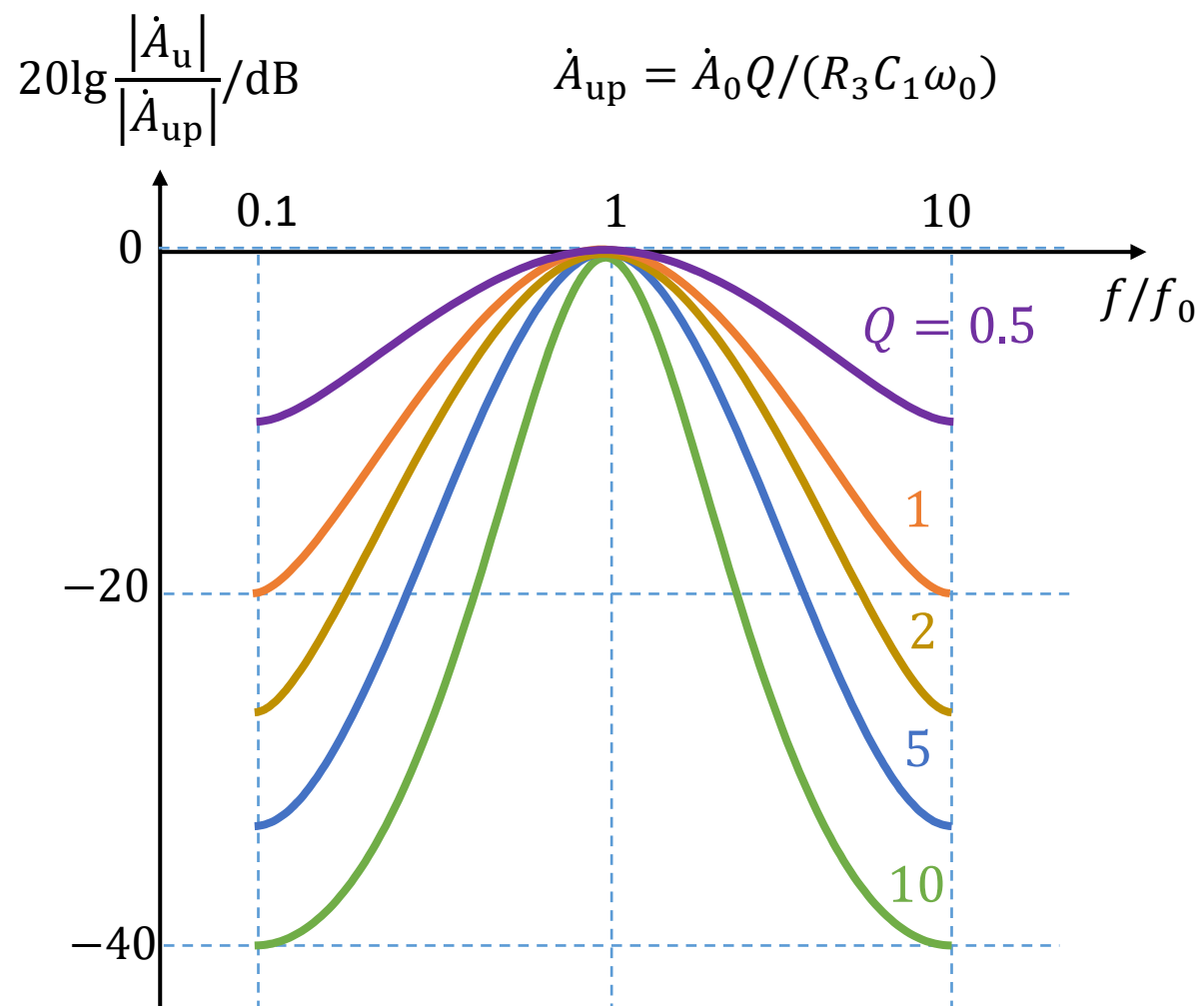
Bandwidth

$$\left\{ \begin{array}{l} f_{\text{pH}} = \left(\sqrt{1 + \frac{1}{4Q^2}} + \frac{1}{2Q} \right) f_0 \\ f_{\text{pL}} = \left(\sqrt{1 + \frac{1}{4Q^2}} - \frac{1}{2Q} \right) f_0 \end{array} \right.$$

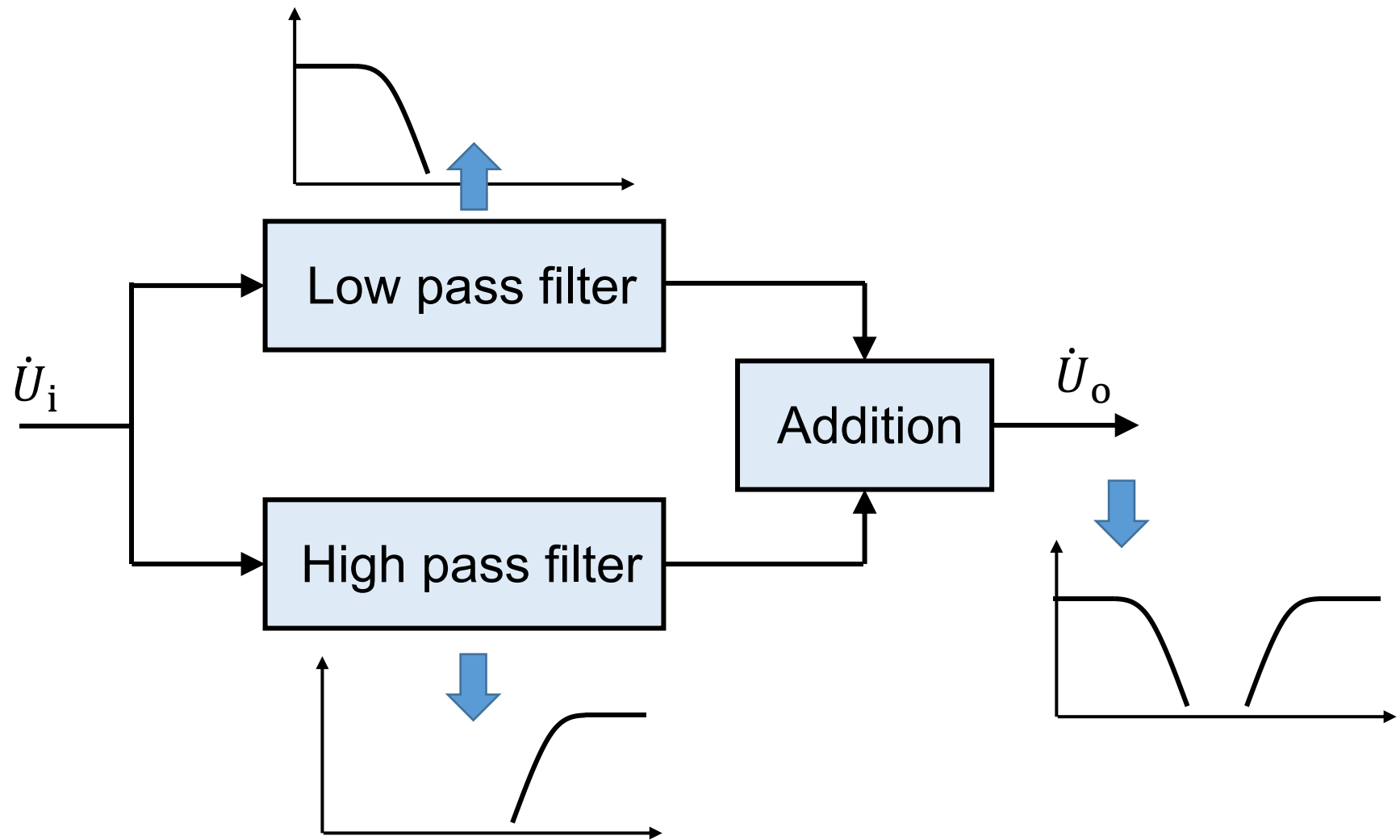
$$f_{\text{bw}} = f_{\text{pH}} - f_{\text{pL}} = \frac{f_0}{Q}$$

Larger Q, smaller bandwidth, better selectivity.

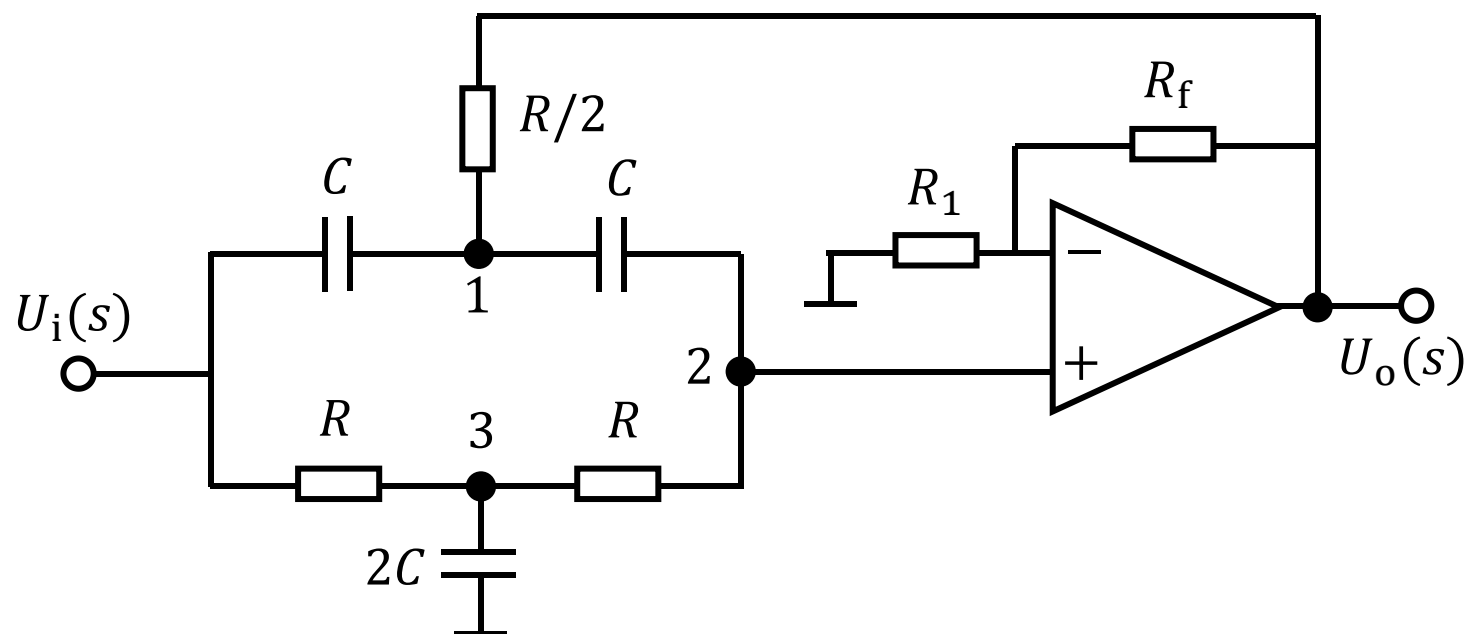
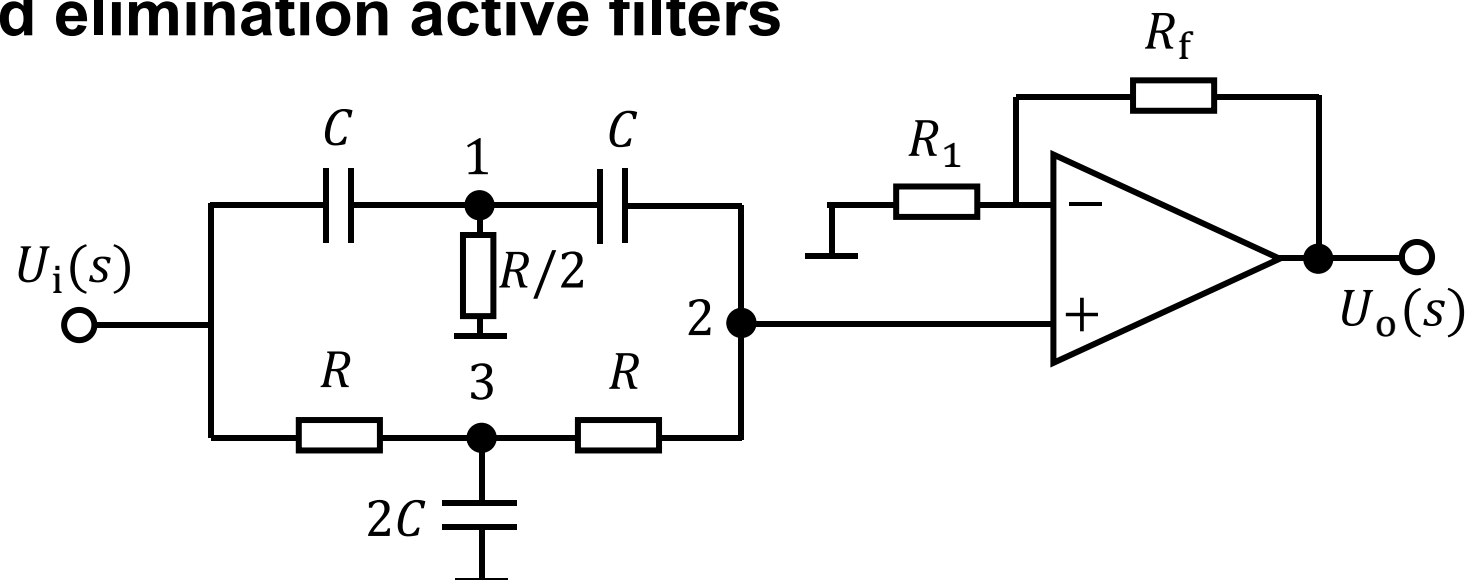
Q值越大，带宽越小，选择性越好

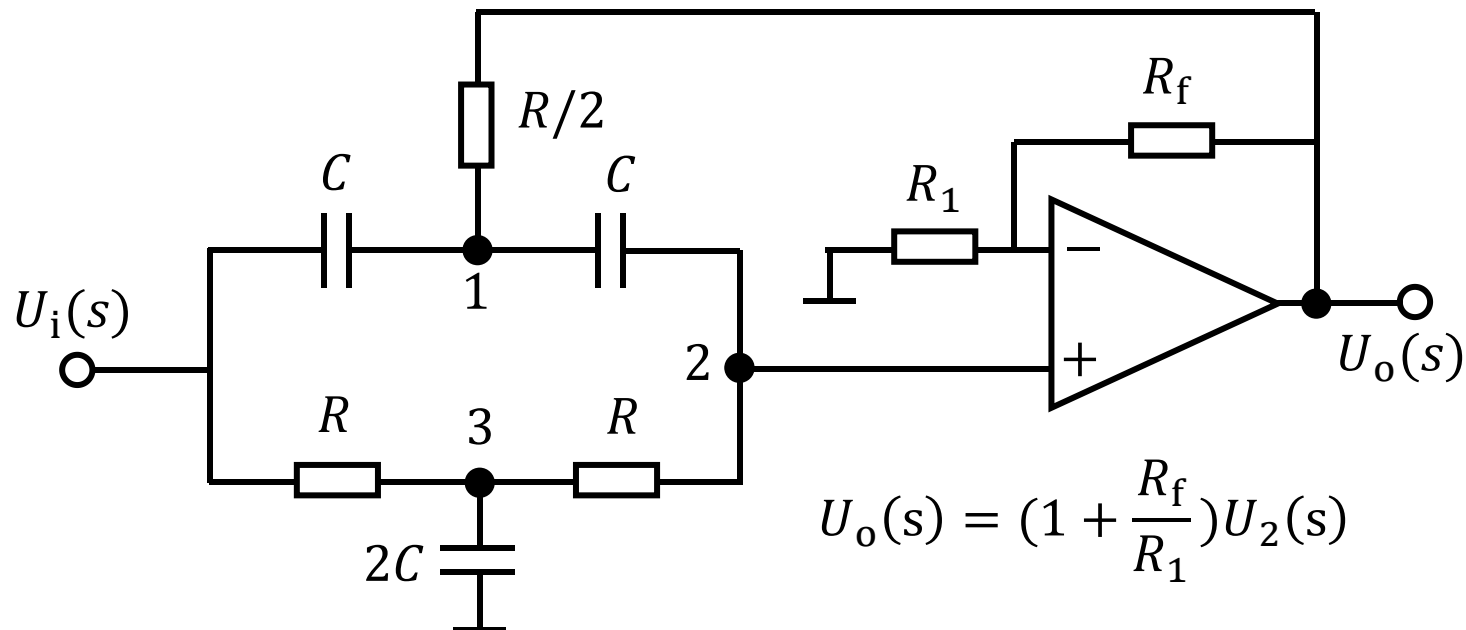


8.3 Band elimination active filters 带阻有源滤波器



Band elimination active filters





Node 1 节点1:
$$\frac{U_i(s) - U_1(s)}{1/sC} + \frac{U_2(s) - U_1(s)}{1/sC} + \frac{U_o(s) - U_1(s)}{R/2} = 0$$

Node 2 节点2:
$$\frac{U_1(s) - U_2(s)}{1/sC} + \frac{U_3(s) - U_2(s)}{R} = 0$$

Node 3 节点3:
$$\frac{U_i(s) - U_3(s)}{R} + \frac{U_2(s) - U_3(s)}{R} + \frac{U_3(s)}{1/2sC} = 0$$

Transfer function 传递函数

$$A_u(s) = \frac{U_o(s)}{U_i(s)}$$

$$= A_0 \frac{s^2 + \omega_0^2}{s^2 + s\omega_0/Q + \omega_0^2}, \quad A_0 = 1 + \frac{R_f}{R_1}$$

$$\left\{ \begin{array}{l} \omega_0 = 2\pi f_0 = \frac{1}{RC} \\ Q = \frac{1}{2(1 - A_0)} \end{array} \right.$$

Frequency response: $s = j\omega$

$$\dot{A}_u = A_u(j\omega) = \dot{A}_0 \frac{\omega_0^2 - \omega^2}{\omega_0^2 - \omega^2 + j\omega\omega_0/Q}, \quad \dot{A}_0 = 1 + \frac{R_f}{R_1}$$

Amplitude frequency characteristic

$$|\dot{A}_u| = \dot{A}_0 \frac{|\omega_0^2 - \omega^2|}{\sqrt{(\omega_0^2 - \omega^2)^2 + \frac{\omega_0^2 \omega^2}{Q^2}}} = \dot{A}_0 \frac{|f_0^2 - f^2|}{\sqrt{(f_0^2 - f^2)^2 + \frac{f_0^2 f^2}{Q^2}}}$$

$$|\dot{A}_u| = \dot{A}_0 \frac{|f_0^2 - f^2|}{\sqrt{(f_0^2 - f^2)^2 + \frac{f_0^2 f^2}{Q^2}}}$$

(a) When $f = 0$, $|\dot{A}_u| = \dot{A}_0$

(b) When $f \rightarrow \infty$, $|\dot{A}_u| \rightarrow \dot{A}_0$

(c) When $f = f_0$, $|\dot{A}_u| = 0$

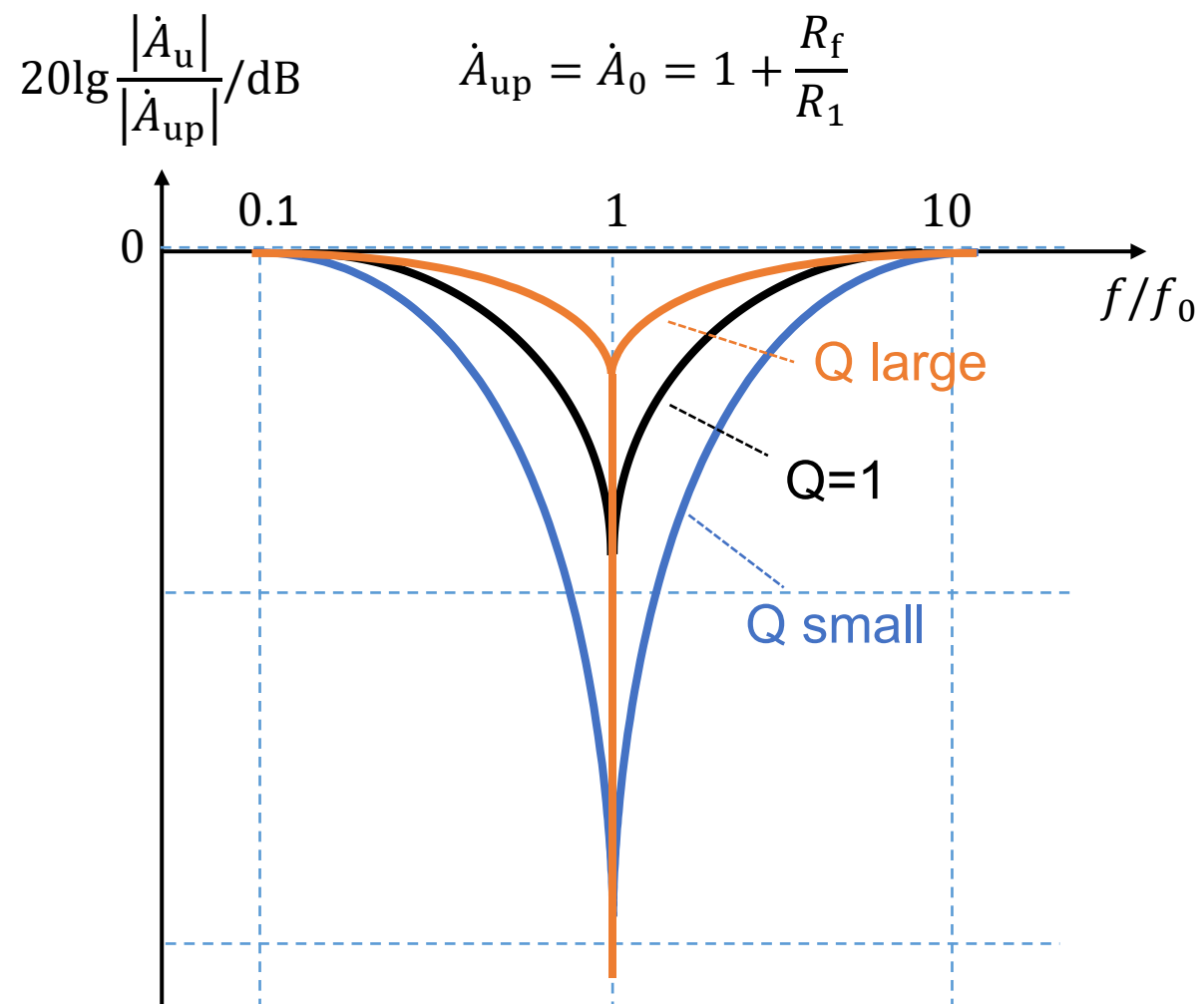
Band elimination filter!

Bandwidth

$$\dot{A}_0 \frac{|f_0^2 - f^2|}{\sqrt{(f_0^2 - f^2)^2 + \frac{f_0^2 f^2}{Q^2}}} = \frac{\dot{A}_0}{\sqrt{2}}$$

$$\left\{ \begin{array}{l} f_{\text{pH}} = \left(\sqrt{1 + \frac{1}{4Q^2}} + \frac{1}{2Q} \right) f_0 \\ f_{\text{pL}} = \left(\sqrt{1 + \frac{1}{4Q^2}} - \frac{1}{2Q} \right) f_0 \end{array} \right.$$

$$f_{\text{bw}} = f_{\text{pH}} - f_{\text{pL}} = \frac{f_0}{Q}$$



Homework 9-1 : refer to blackboard